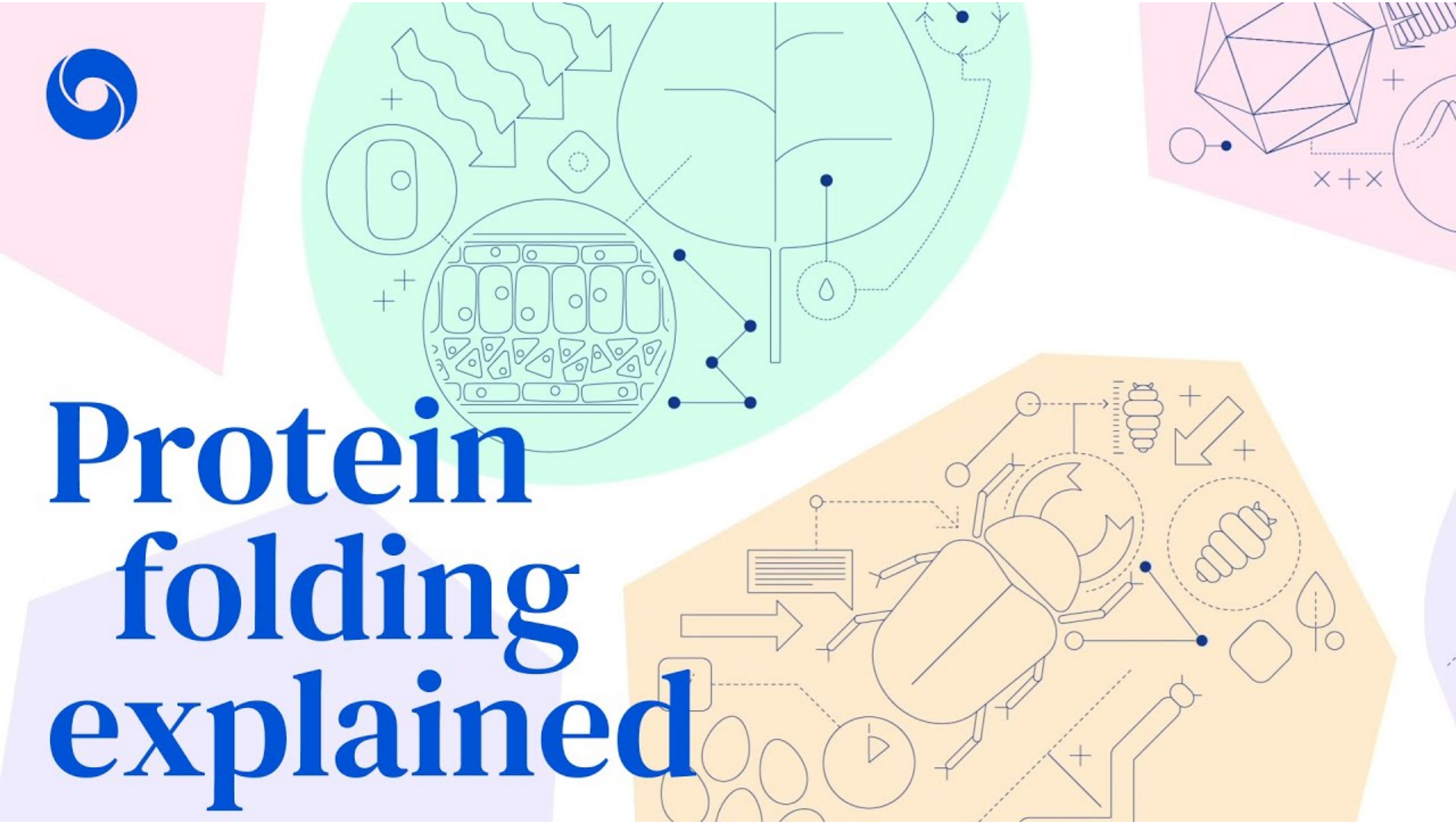
Introduction to ML

Nipun Batra

Machine Learning Applications

Protein folding explained: https://www.youtube.com/watch?v=KpedmJdrTpY



Machine Learning Applications

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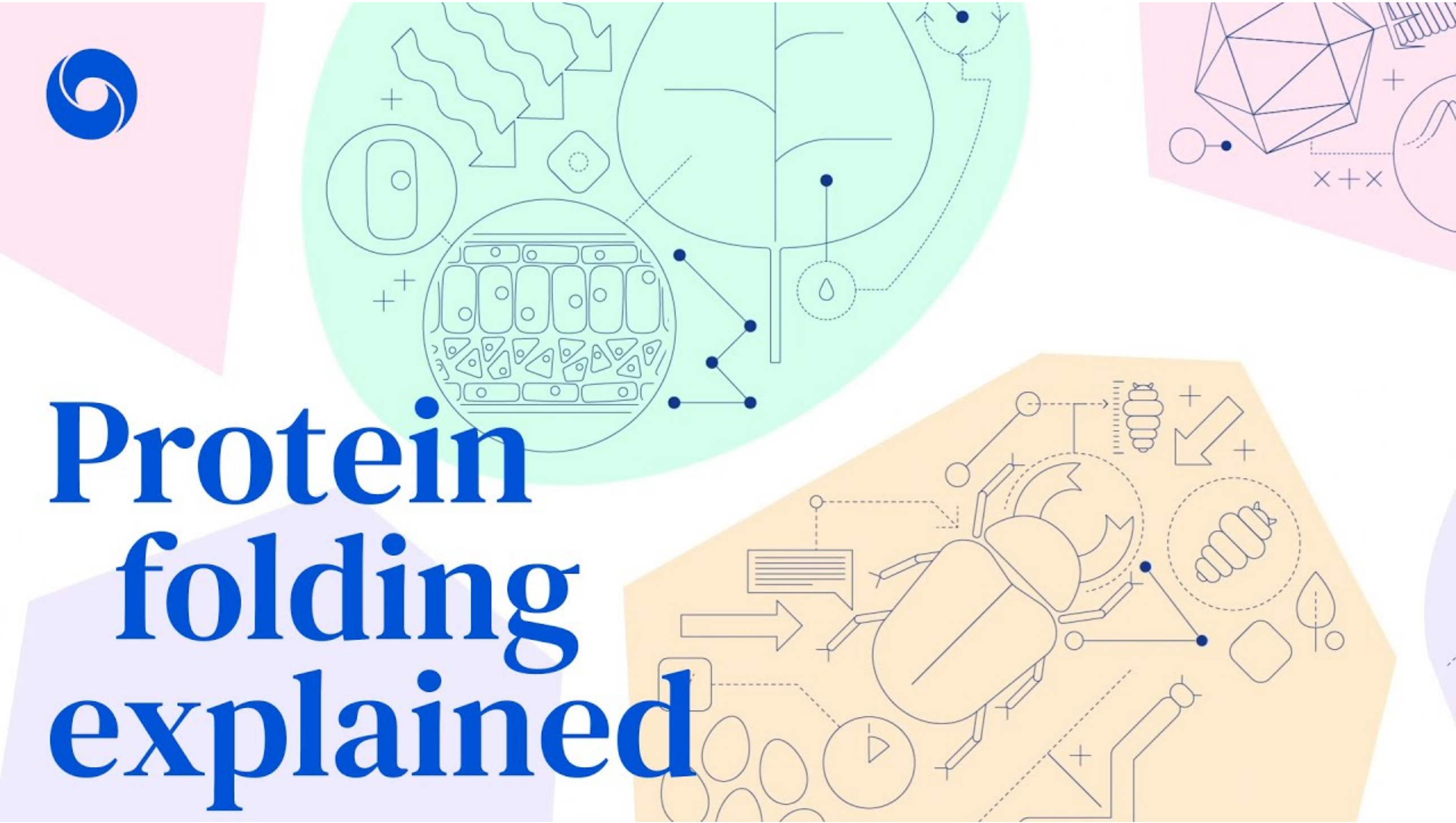
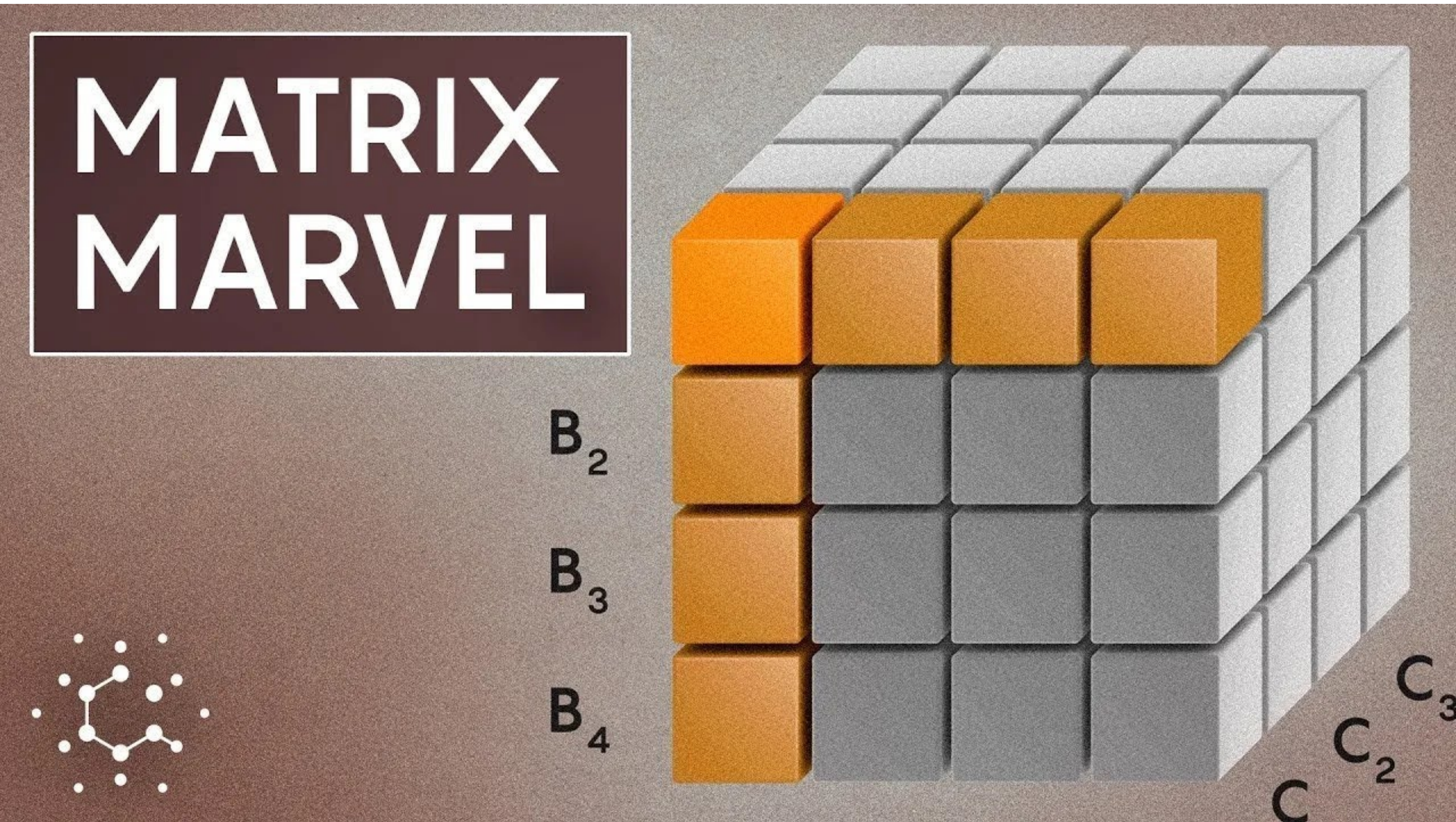


Machine Learning Applications

How AI Discovered a Faster Matrix Multiplication Algorithm https://www.youtube.com/watch?v=fDAPJ7rvcUw



Machine Learning Applications

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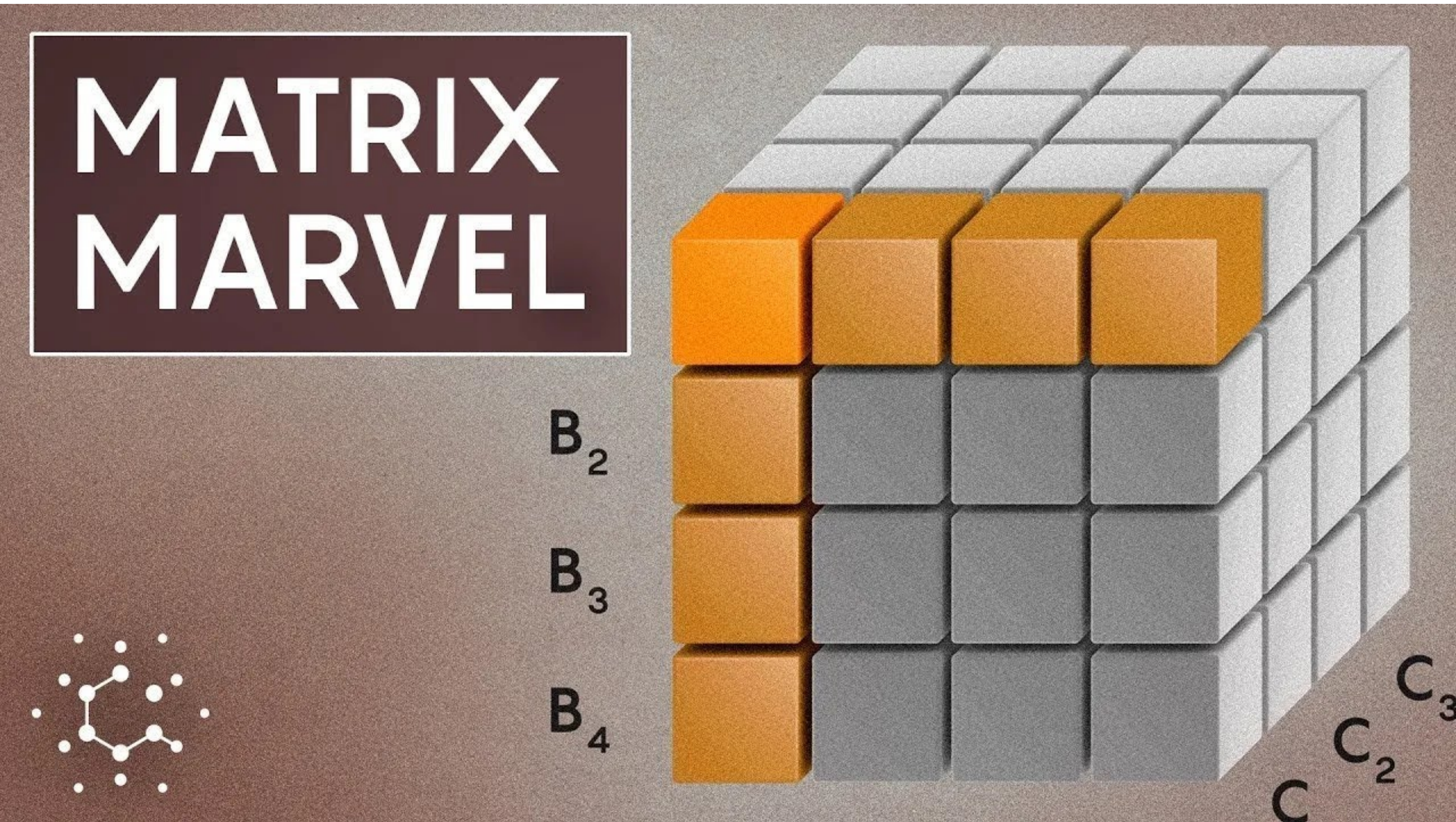

Machine Learning Applications

(Already outdated: Devin) https://www.youtube.com/watch?v=fjHtjT7GO1c

Machine Learning Applications

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Machine Learning Applications

6 weeks of Claude Code:https://news.ycombinator.com/item?id=44746621

▲ lukaslalinsky 3 days ago | next [–]

I have about two weeks of using Claude Code and to be honest, as a vibe coding skeptic, I was amazed. It has a learning curve. You need to learn how to give it proper context, how to chunk up the work, etc. And you need to know how to program, obviously. Asking it to do something you don't know how to do, that's just asking for a disaster. I have more than 25 years of experience, so I'm confident with anything Claude Code will try to do and can review it, or stop and redirect it. About 10-15 years ago, I was dreaming about some kind of neural interface, where I could program without writing any code. And I realized that with Claude Code, it's kind of here.

A couple of times I hit the daily limits and decided to try Gemini CLI with the 2.5 pro model as a replacement. That's not even comparable to Claude Code. The frustration with Gemini is just not worth it.

I couldn't imagine paying >100$/month for a dev tool in the past, but I'm seriously considering upgrading to the Max plans.

reply

▲ giancarlostoro 3 days ago | parent | next [–]

If you are a Senior Developer, who is comfortable giving a Junior tips, and then guiding them to fixing them (or just stepping in for a brief moment and writing where they missed something) this is for you. I'm hearing from Senior devs all over thought, that Junior developers are just garbage at it. They product slow, insecure, or just outright awful code with it, and then they PR the code they don't even understand.

For me the sweet spot is for boilerplate (give me a blueprint of a class based on a description), translate a JSON for me into a class, or into some other format. Also "what's wrong with this code? How would a Staff Level Engineer white it?" those questions are also useful. I've found bugs before hitting debug by asking what's wrong with the code I just pounded on my keyboard by hand.

reply

▲ channel_t 3 days ago | root | parent | next [–]

Yes, can confirm that as a senior developer who has needed to spend huge amounts of time reviewing junior code from off-shore contractors with very detailed and explicit instructions, dabbling in agentic LLM coding tools like Claude Code has felt like like a gift from heaven.

I also have concerns about said junior developers wielding such tools, because yes, without being able to supply the right kind of context and being able to understand the difference between a good solution and a bad solution, they will produce tons of awful, but technically working code.

reply

▲ willsmith72 3 days ago | root | parent | next [–]

Totally agree with the off-shore component of this. I'm already going to have to break a task down into clear detail and resolve any anticipated blocker myself upfront to avoid multi-timezone multi-day back and forth.

Now that I'm practiced at that, the off-shored part is no longer valuable

reply

▲ sitkack 3 days ago | root | parent | next [–]

The unemployment in India is going to be catastrophic. Geopolitical.

reply

Machine Learning Applications

(AI for weather) https://www.youtube.com/watch?v=-KFO0pES-zQ

AI is Transforming Weather Forecasting

Machine Learning Applications

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Machine Learning Applications

(AI for Disaster) https://www.youtube.com/watch?v=ET04pDj-RvM

help save lives
and improve resilience worldwide

Machine Learning Applications

(AI for Disaster) https://www.youtube.com/watch?v=ET04pDj-RvM

help save lives
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Saving The Planet - One Watt A time

Bidgeley: https://www.youtube.com/@bidgely1905

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ML for Farm

Farmbeat: https://www.youtube.com/watch?v=pDgjOHY7sMI

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ML for Health

ML for Health

ML for Education

https://www.youtube.com/watch?v=IvXZCocyU_M

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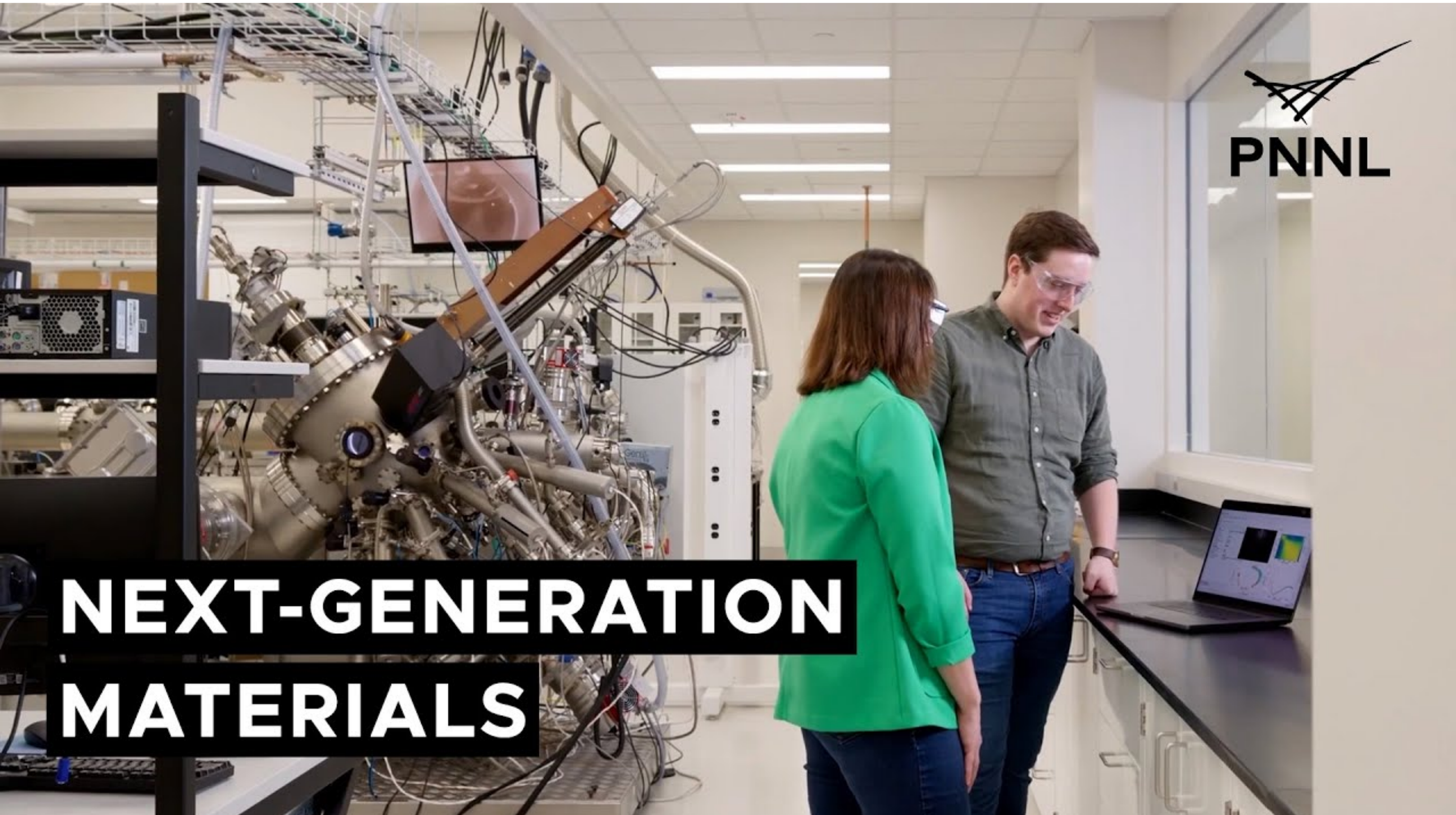
ML for Materials

https://www.youtube.com/watch?v=yRVyehD5lhM



ML for Materials

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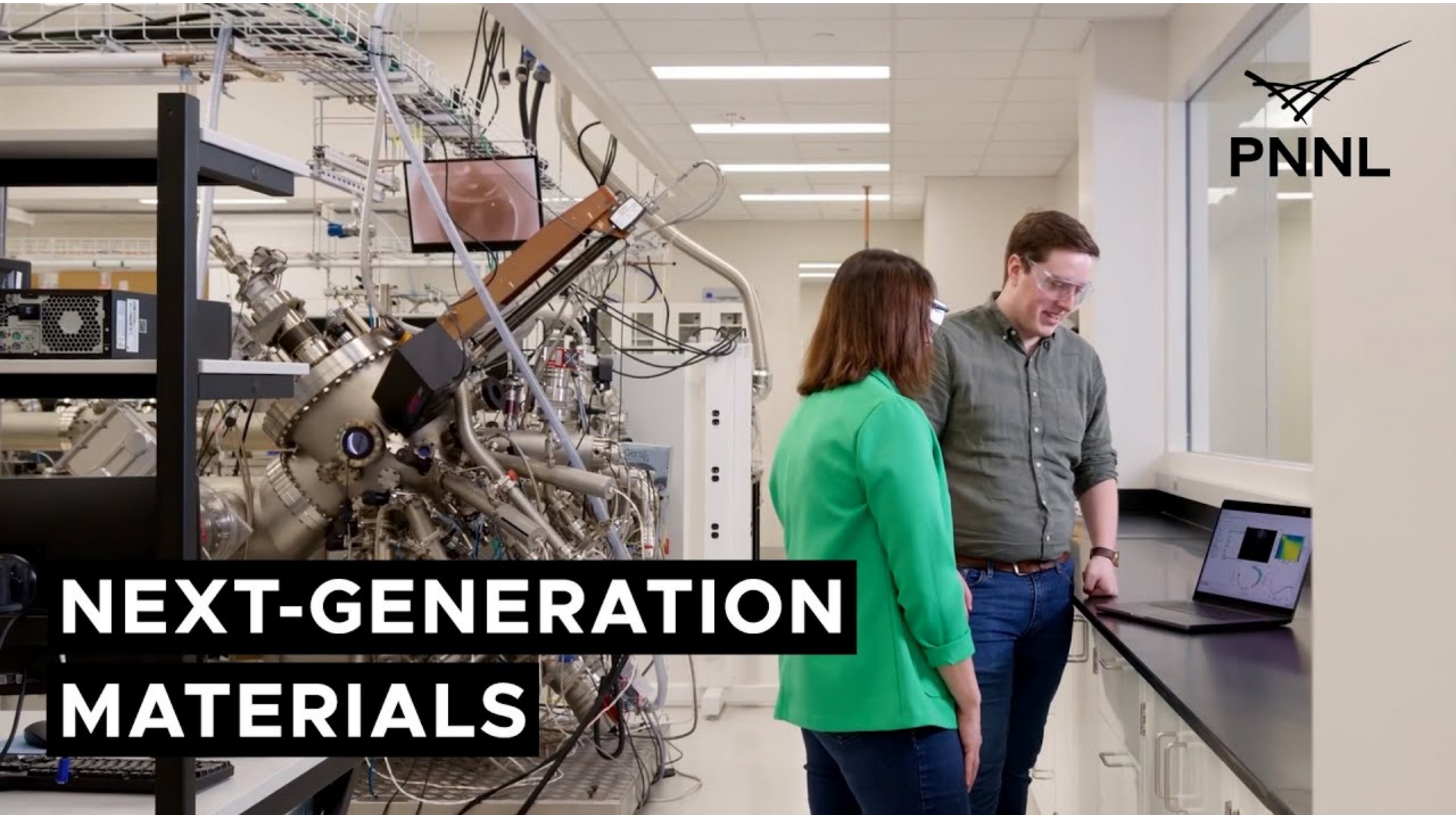
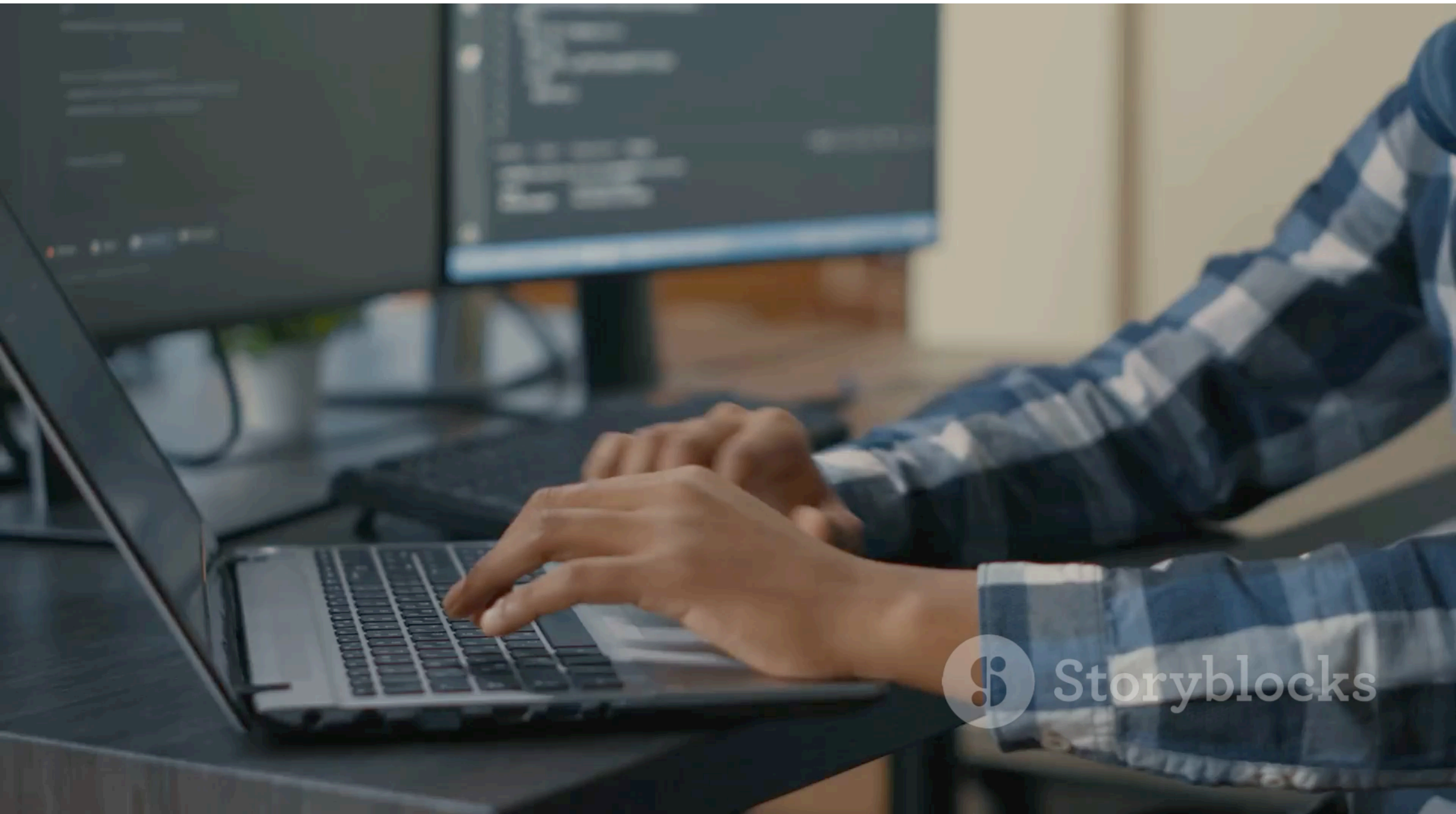


ML for Creativity



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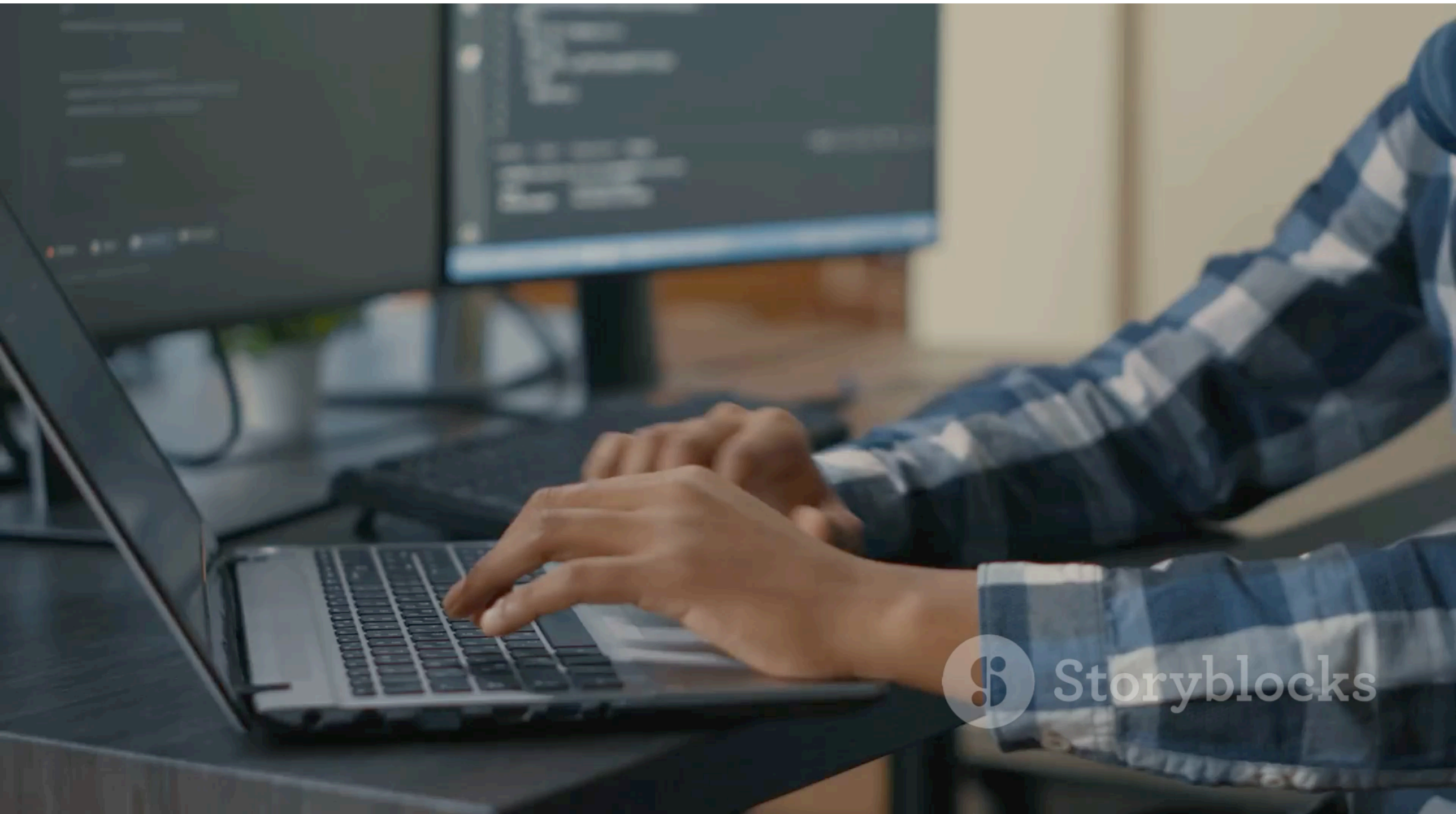
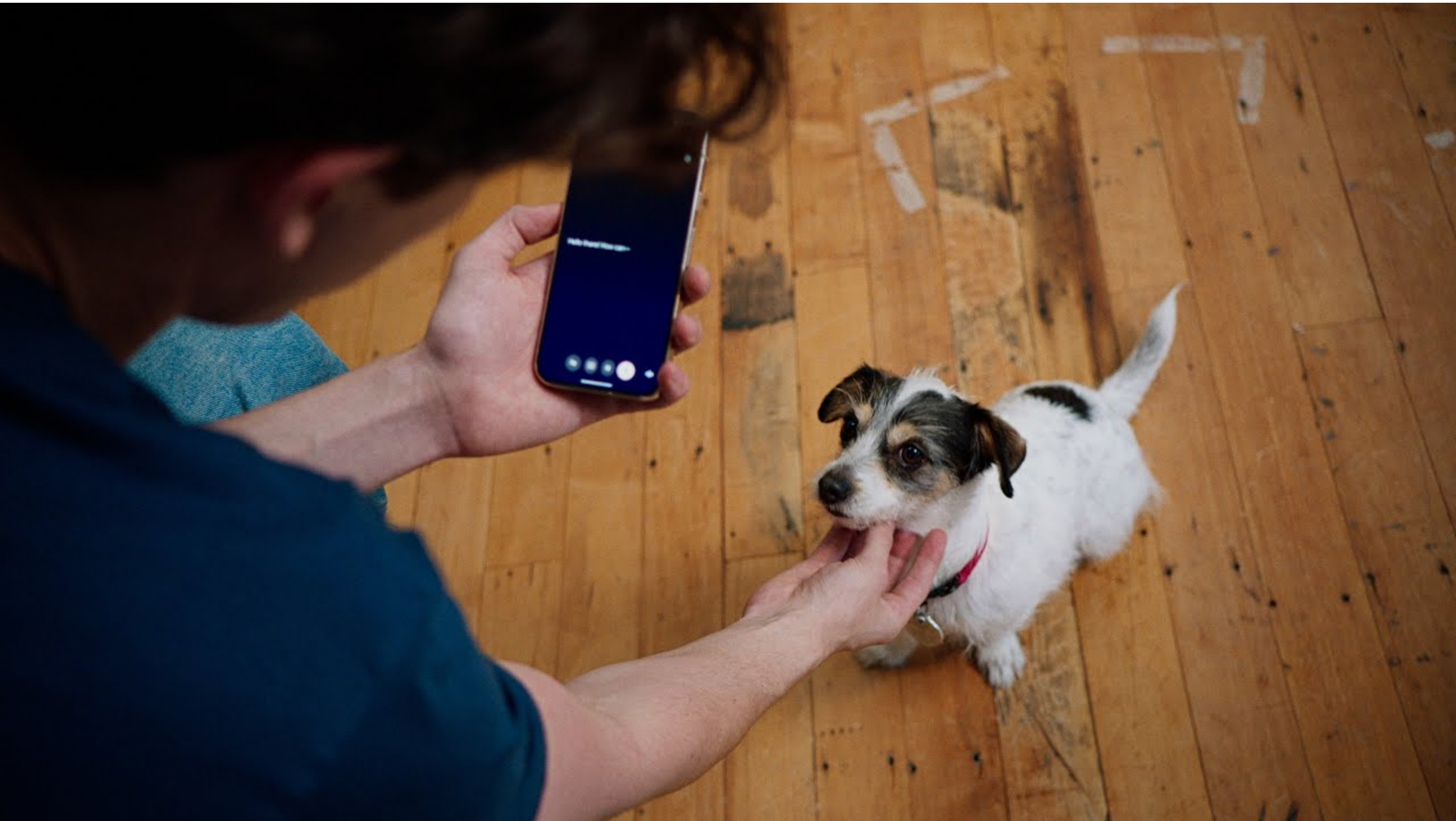


Machine Learning Applications

Demo: https://nipunbatra.github.io/ml-teaching/notebooks/text_to_image.html

Machine Learning Applications

Universal AI Assistant: https://www.youtube.com/watch?v=JcDBFAm9PPI



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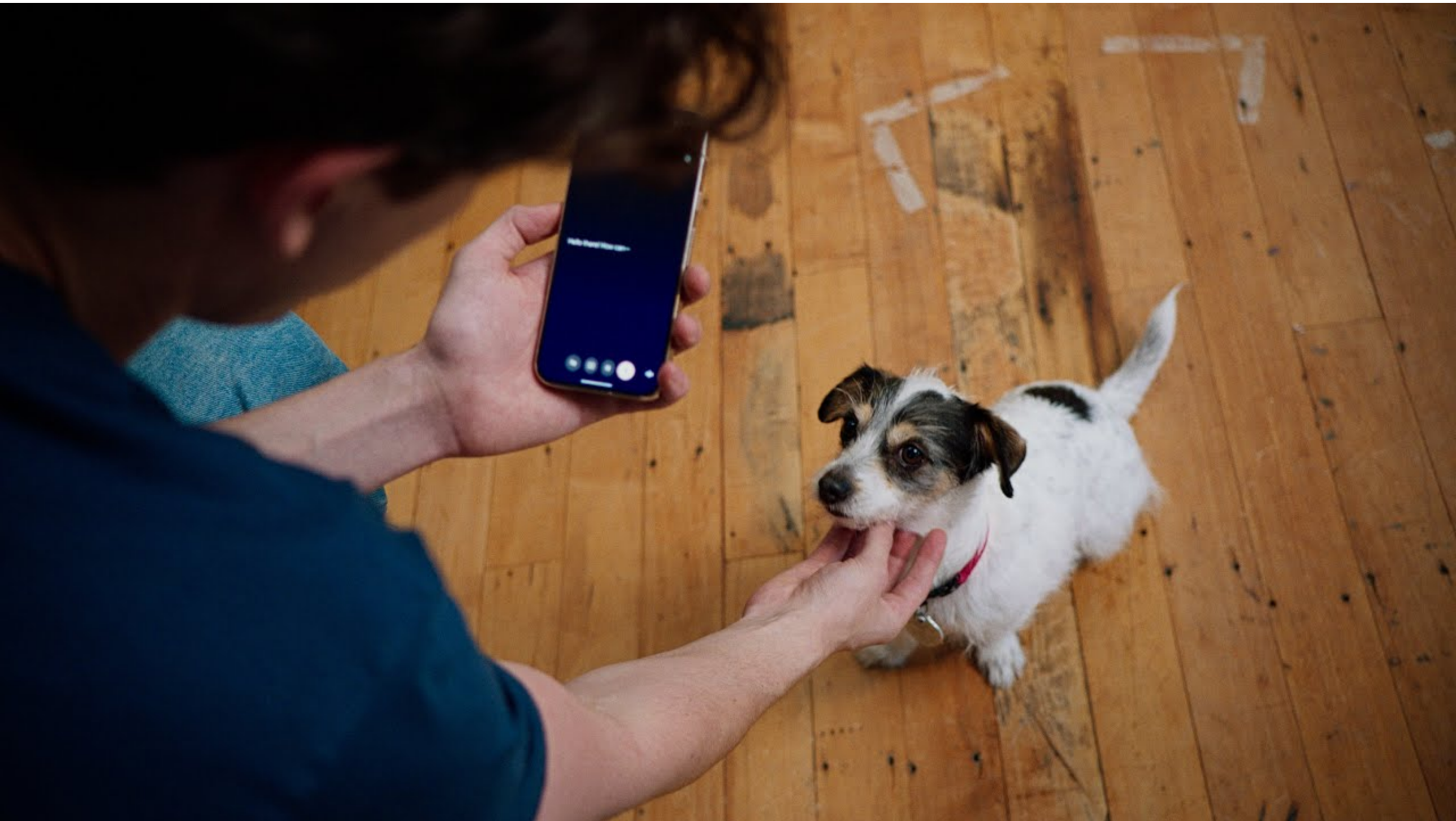

Machine Learning Applications

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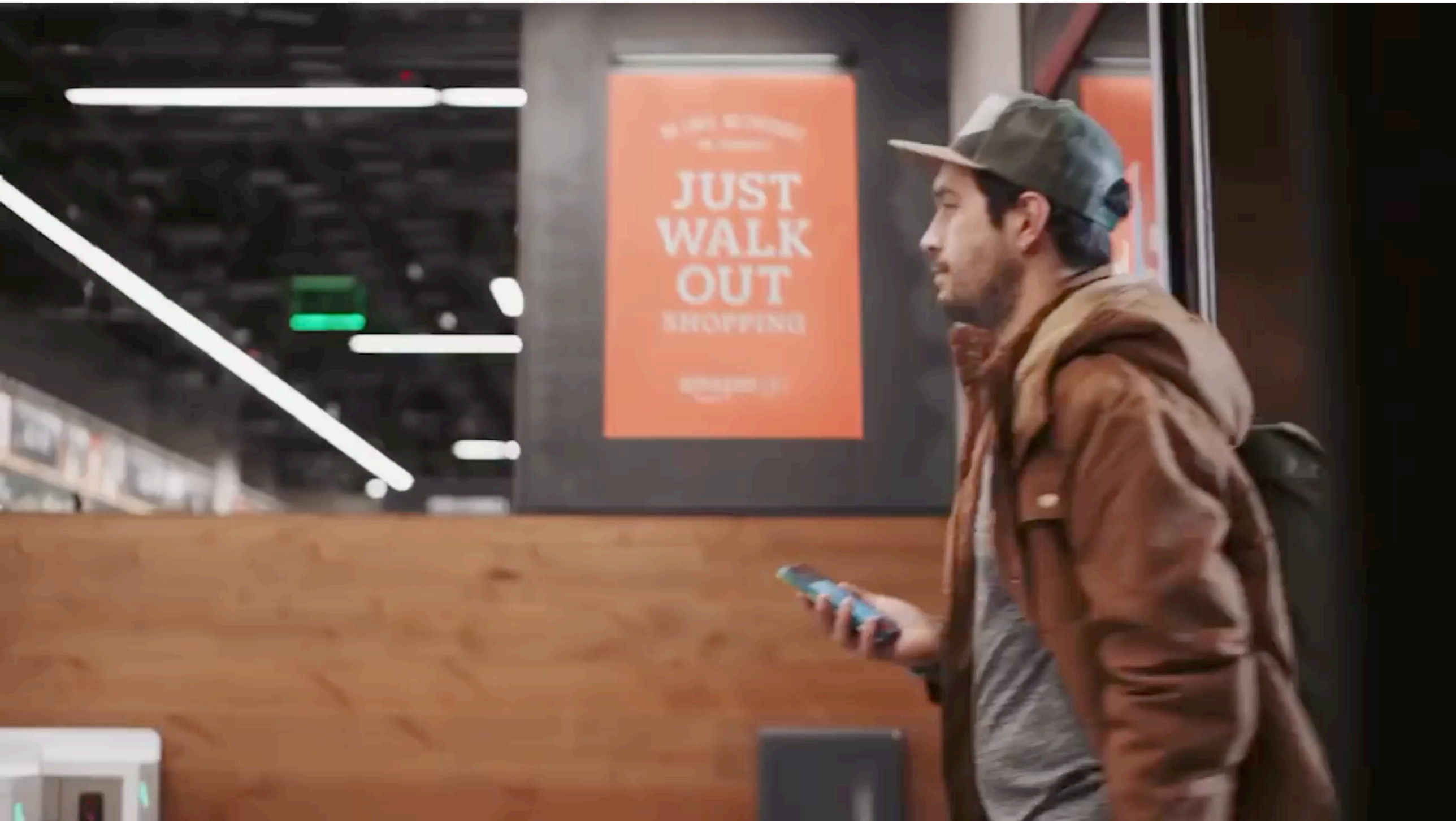
The Long Wait …

Machine Learning Applications

https://www.youtube.com/watch?v=NrmMk1Myrxc



Machine Learning Applications

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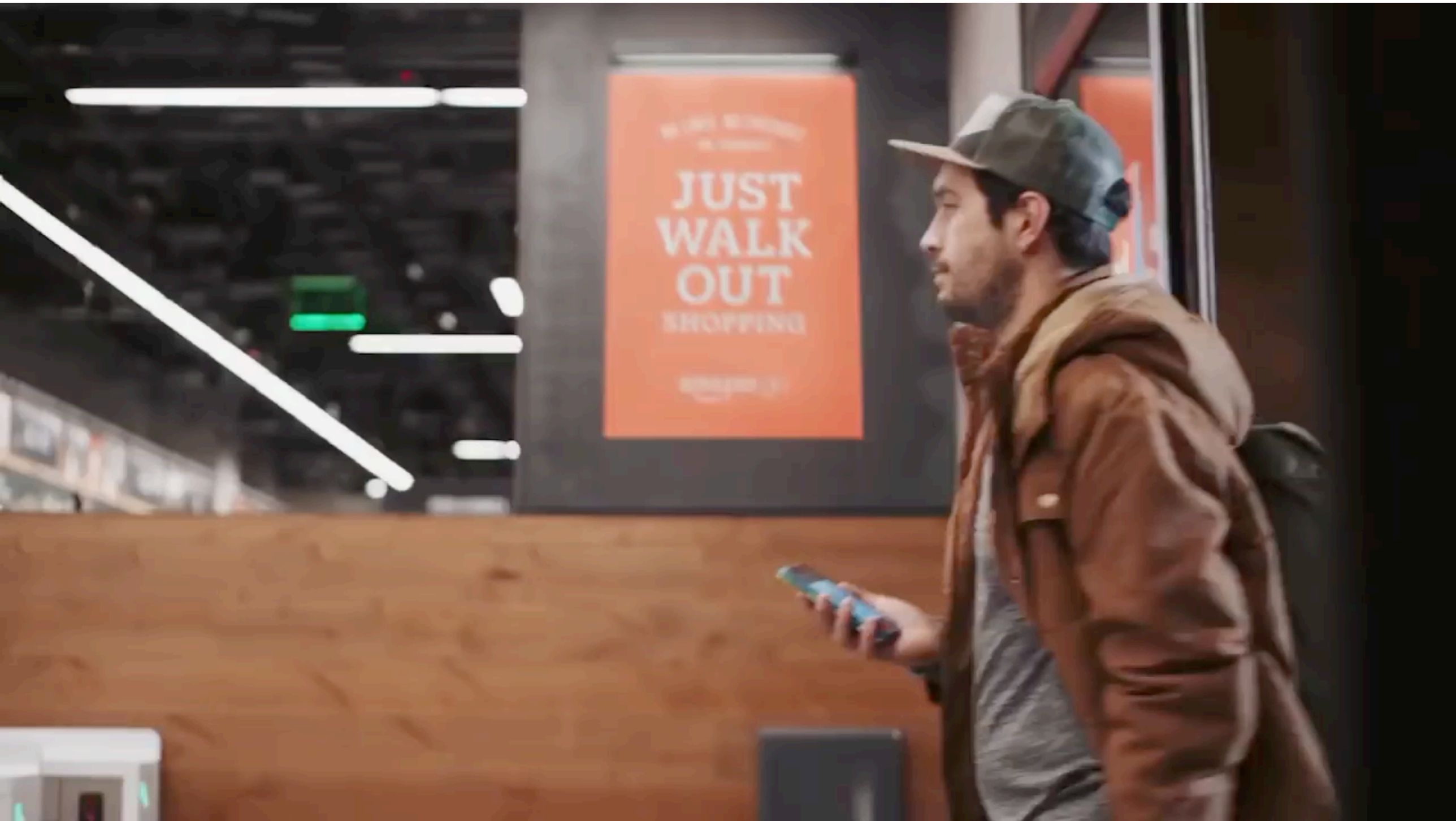

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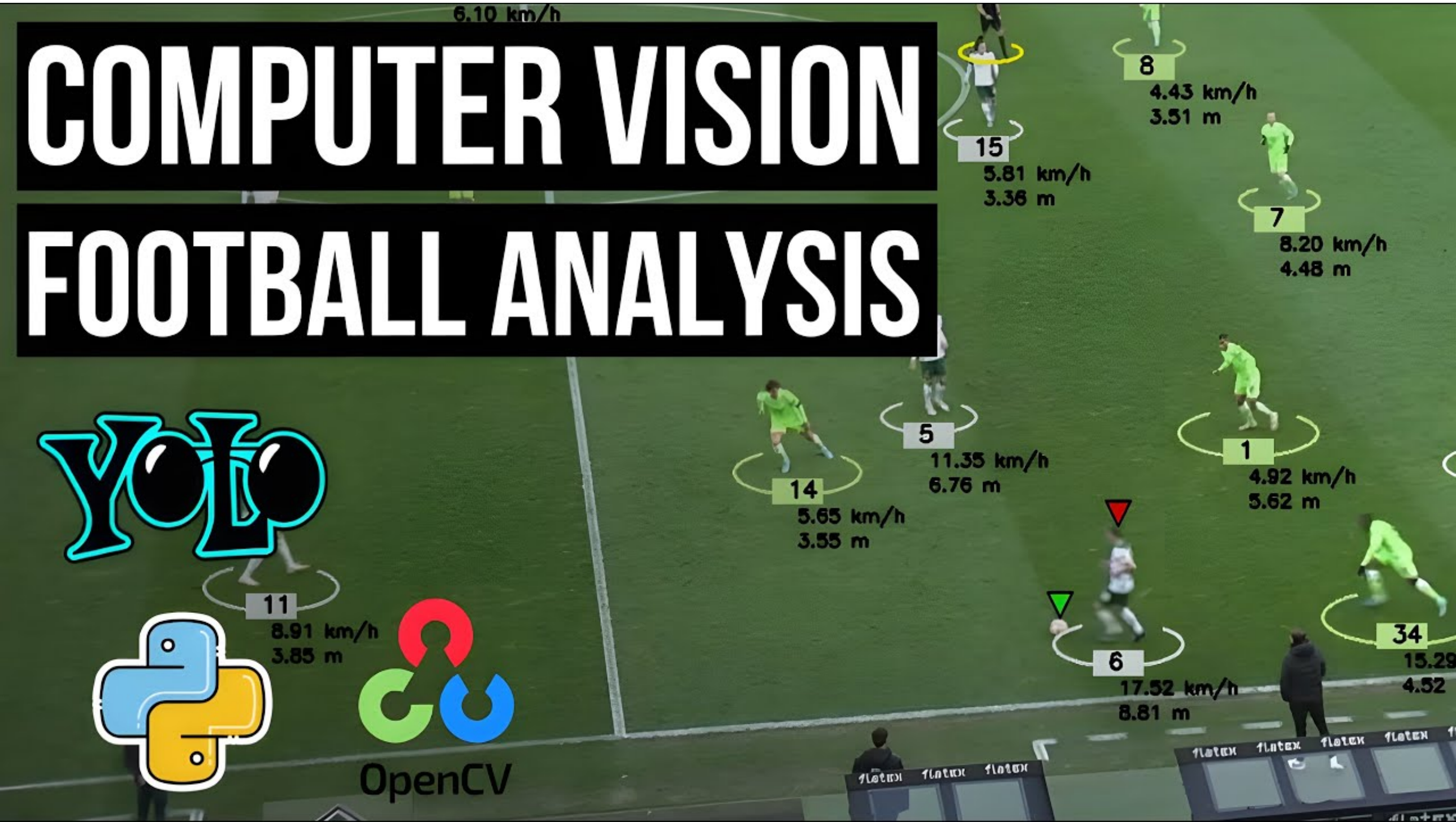

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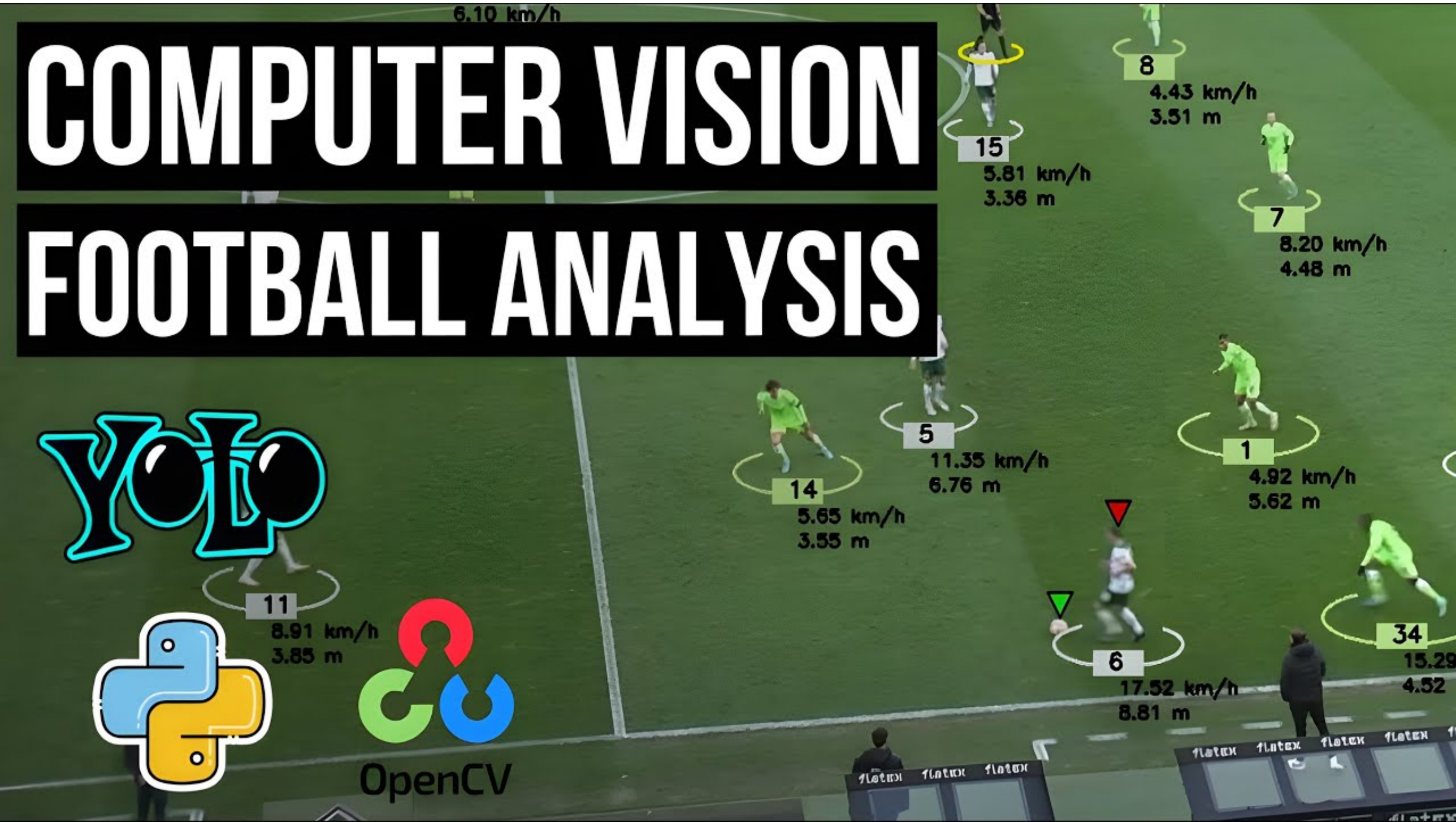
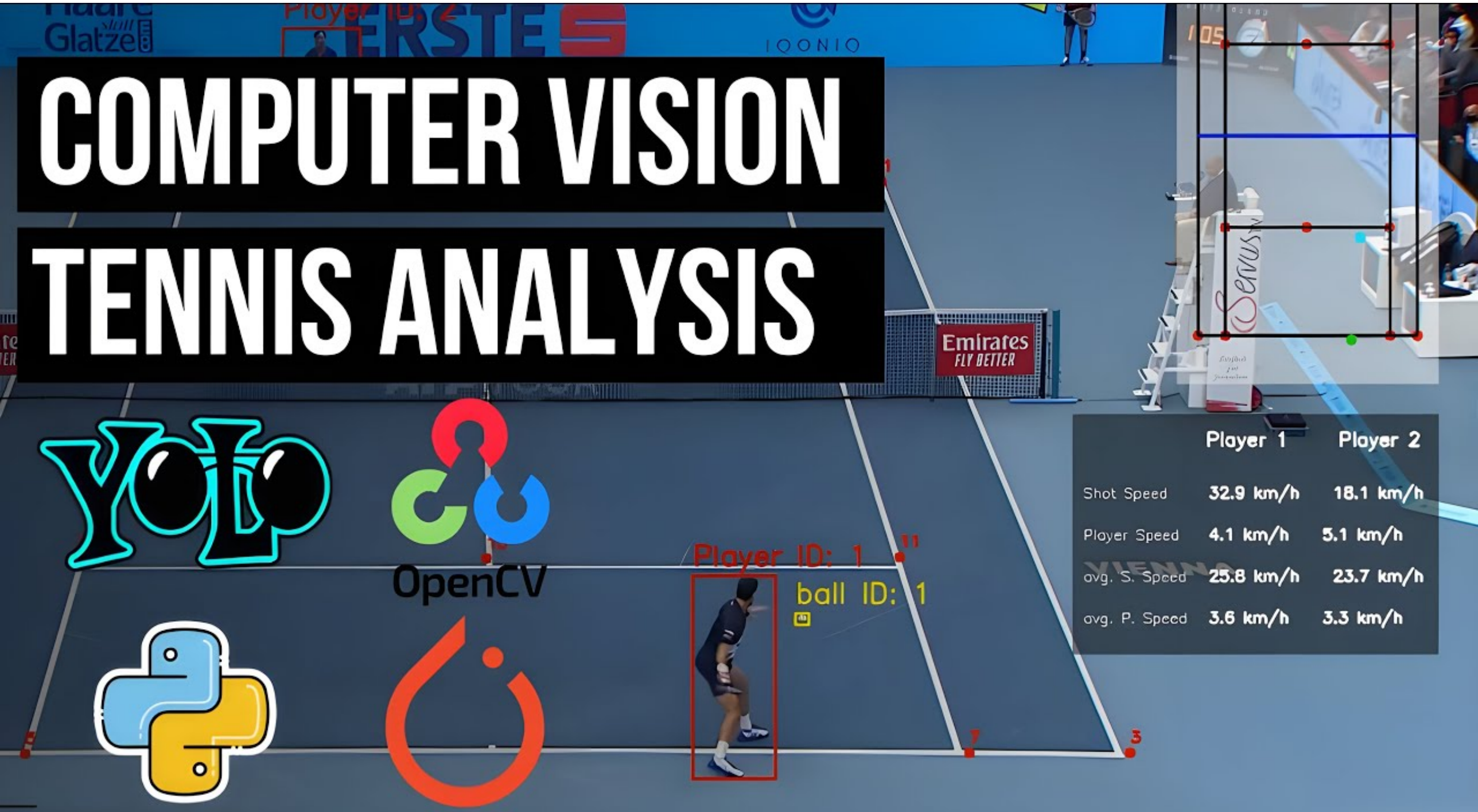


Machine Learning Applications



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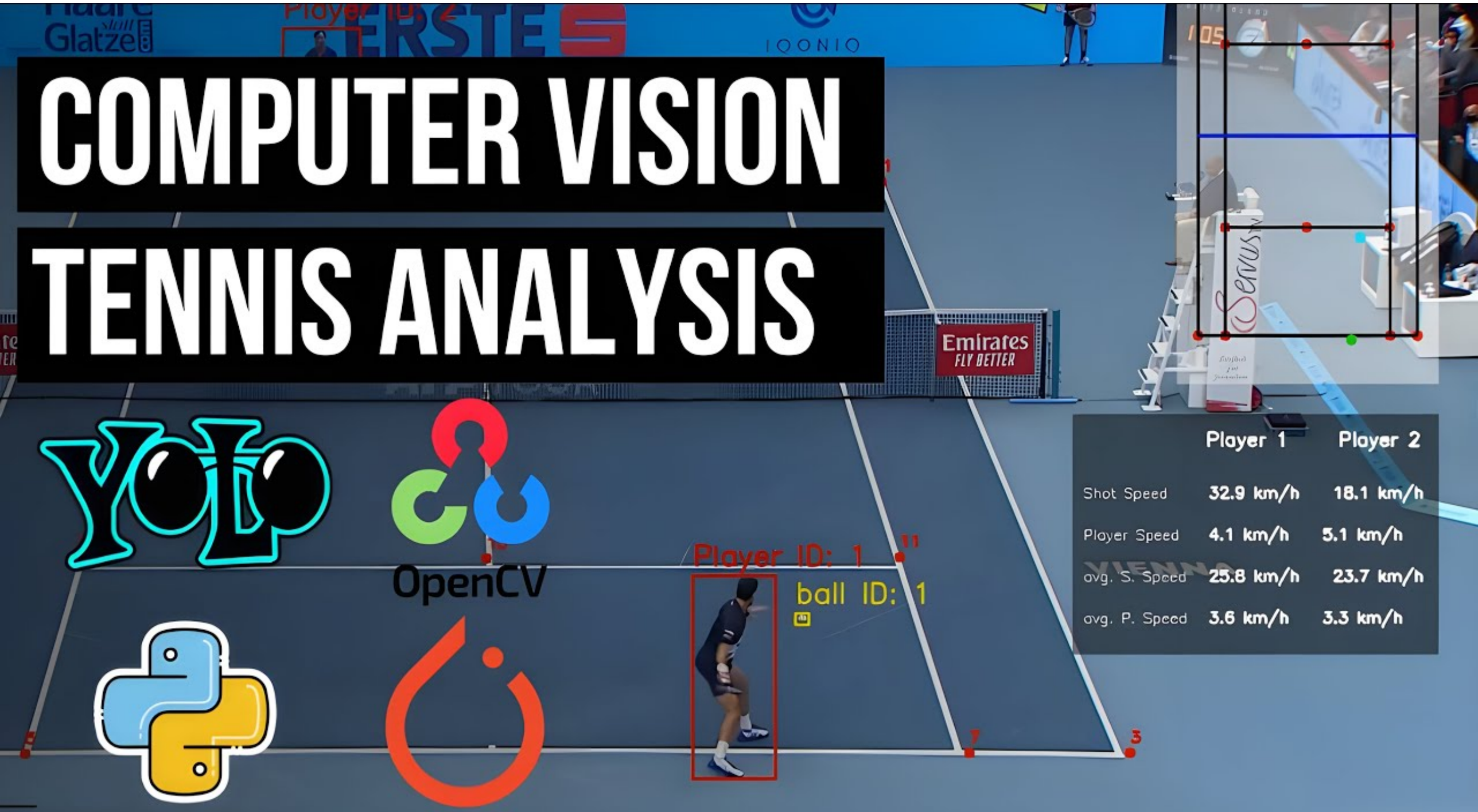
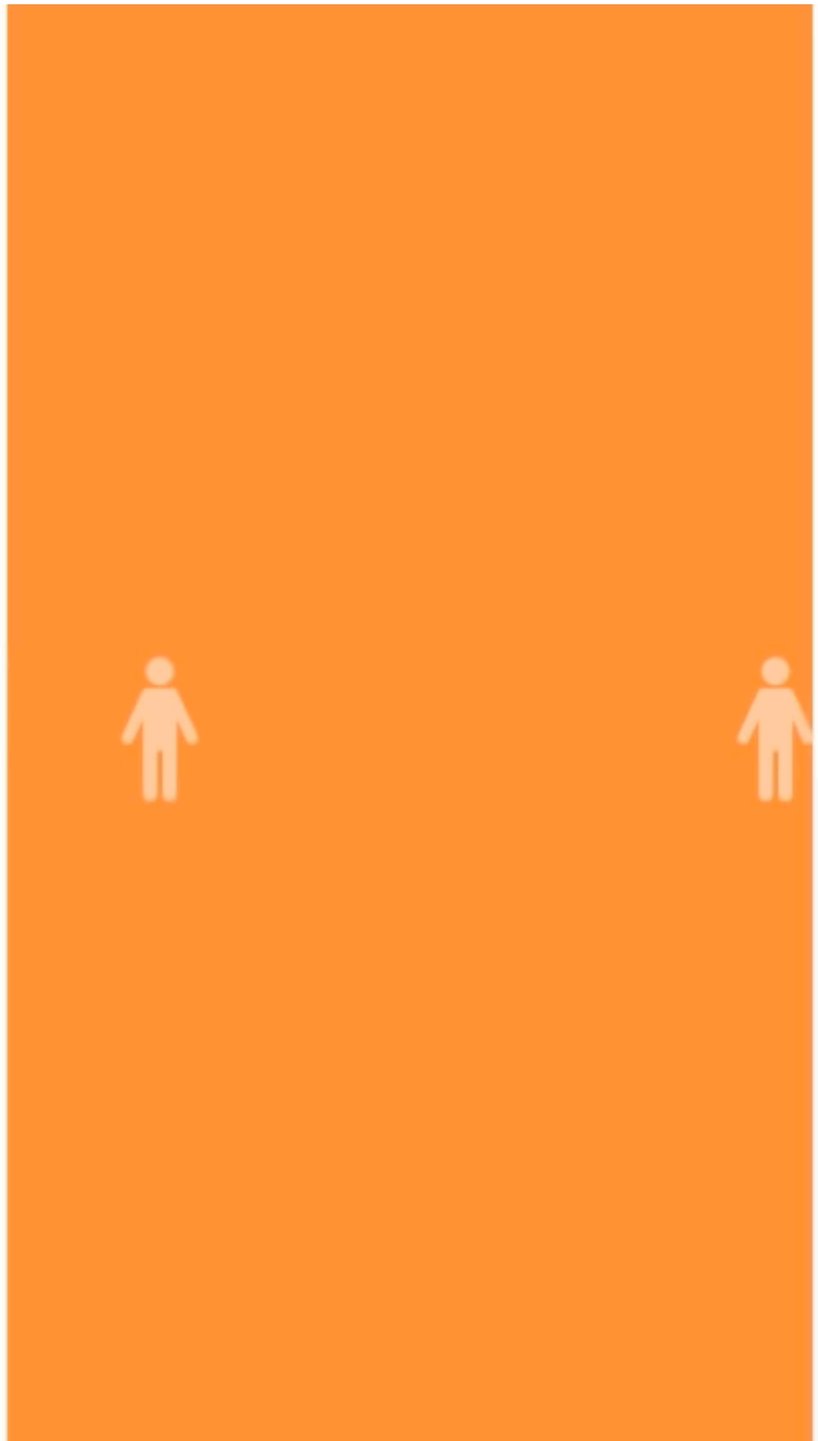


Machine Learning Applications

Demo: https://nipunbatra.github.io/ml-teaching/notebooks/object-detection-segmentation.html

Machine Learning Applications

Poverty detection using satellite images: https://www.youtube.com/watch?v=DafZSeIGLNE



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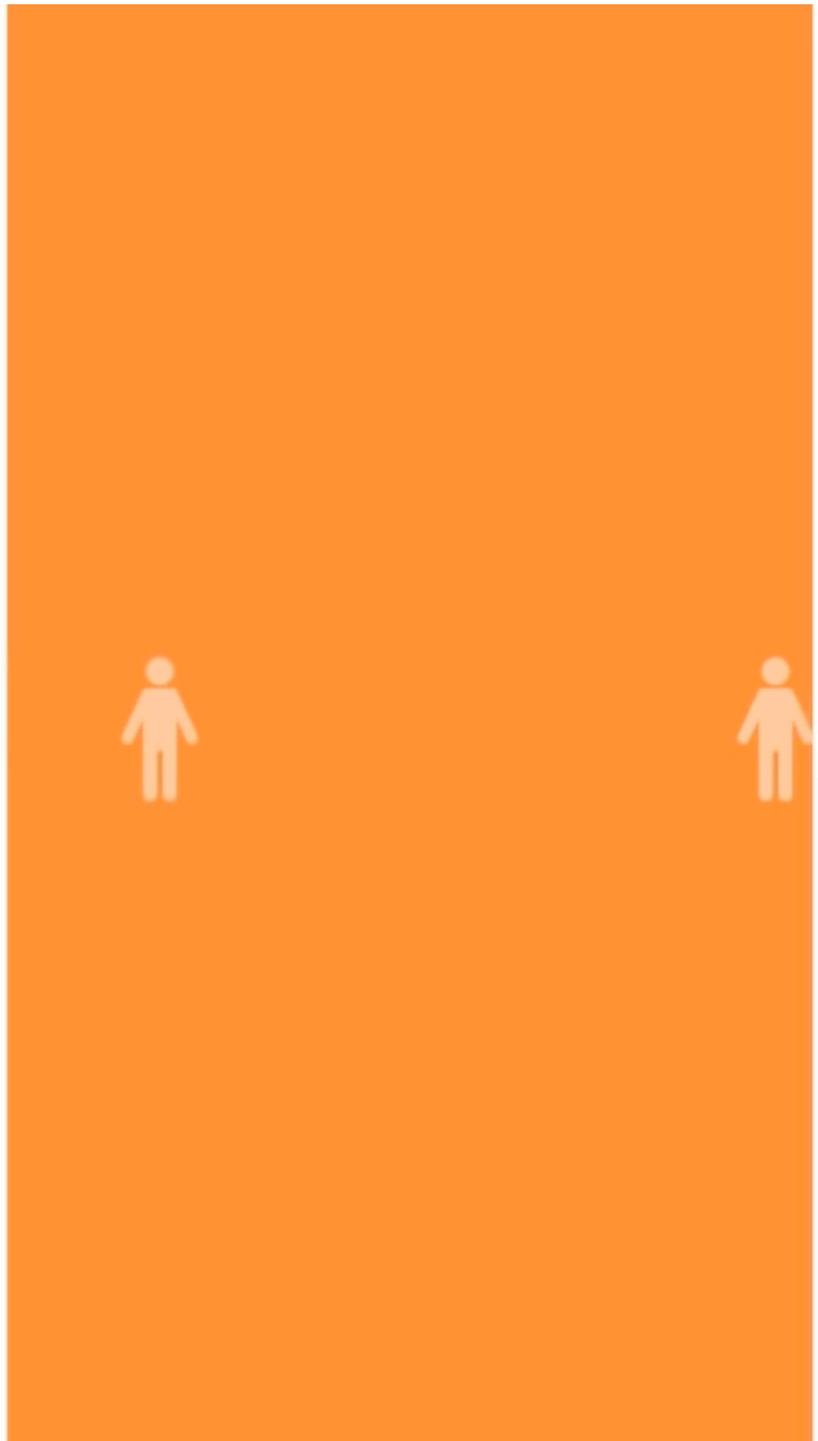

Self Driving Car

Waymo self-driving car: https://www.youtube.com/@Waymo

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Swaayat Robotics: https://www.youtube.com/watch?v=0rFuFVHOwG8

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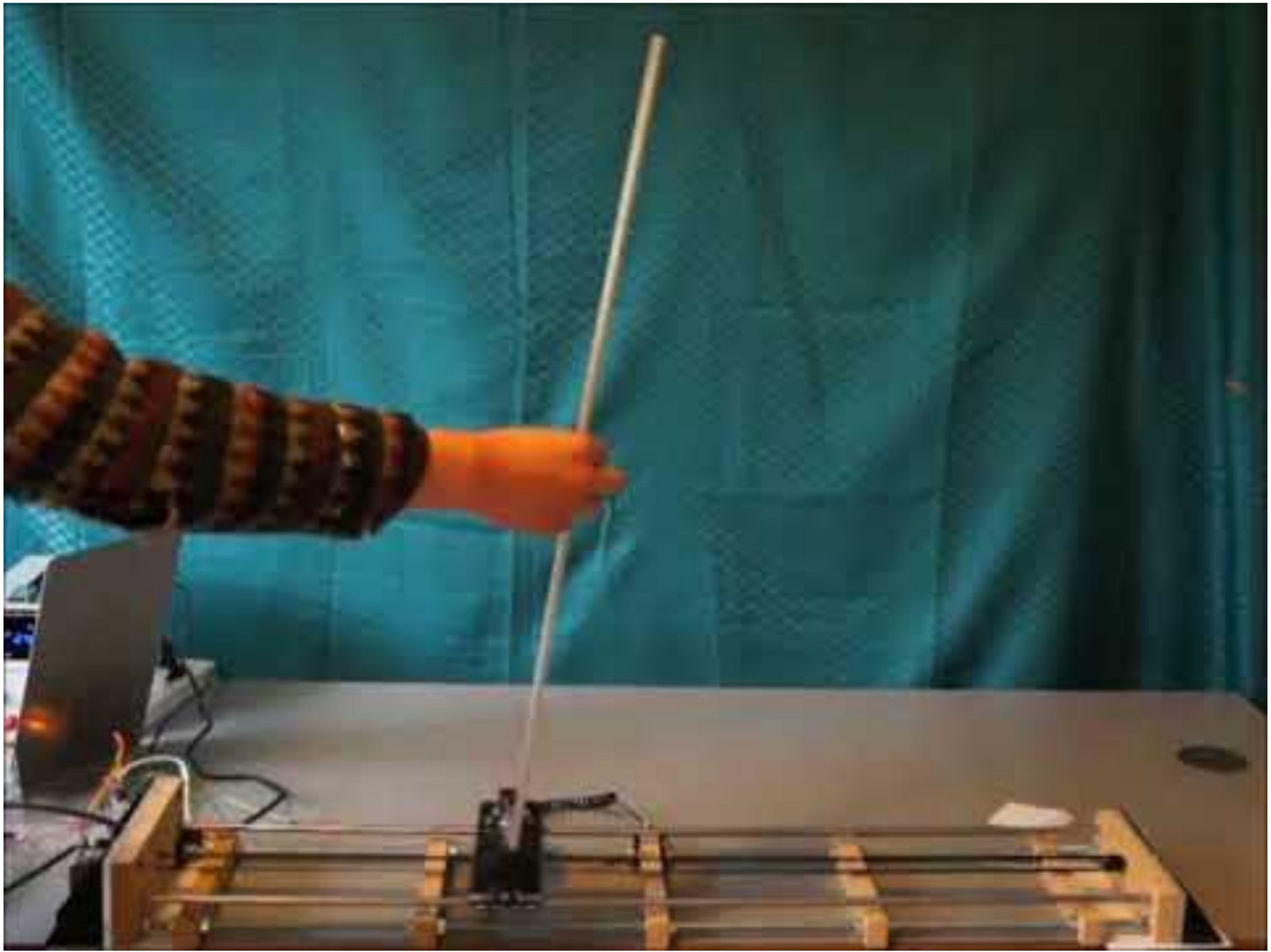
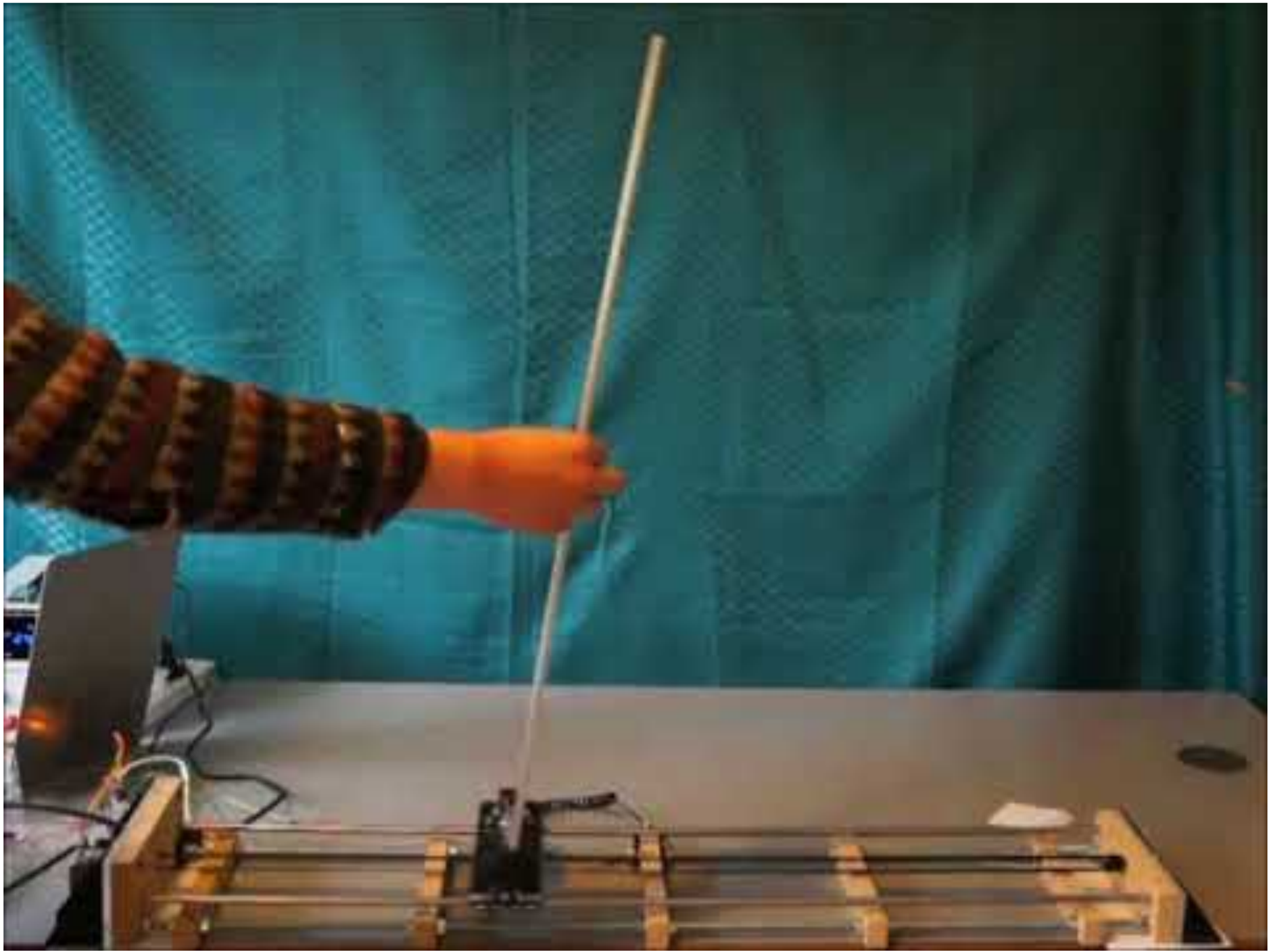
Self Driving Car

Cart Pole RL: https://youtube.com/watch?v=5Q14EjnOJZc



Self Driving Car

Cart Pole RL: https://youtube.com/watch?v=5Q14EjnOJZc



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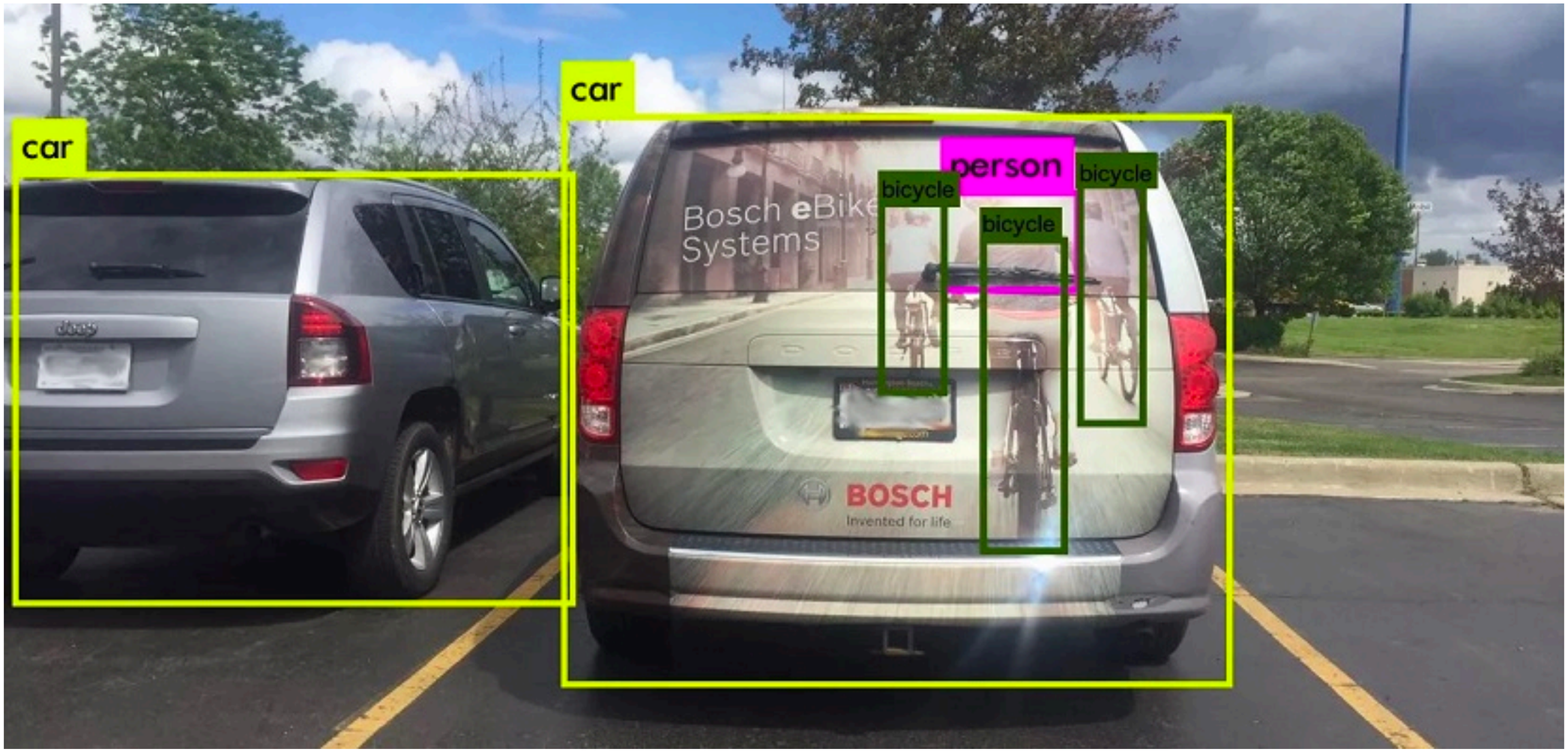

Courtesy: Cognata

Machine Learning Applications

https://www.youtube.com/watch?v=PDKhUknuQDg

Machine Learning Applications

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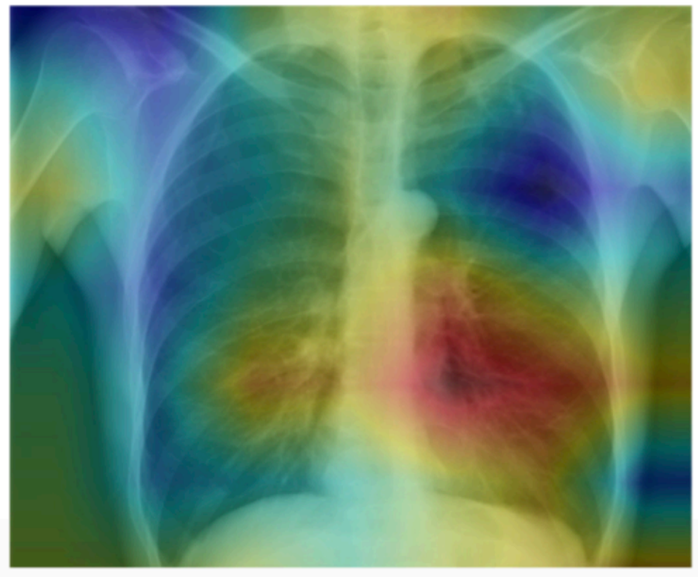
Machine Learning Applications

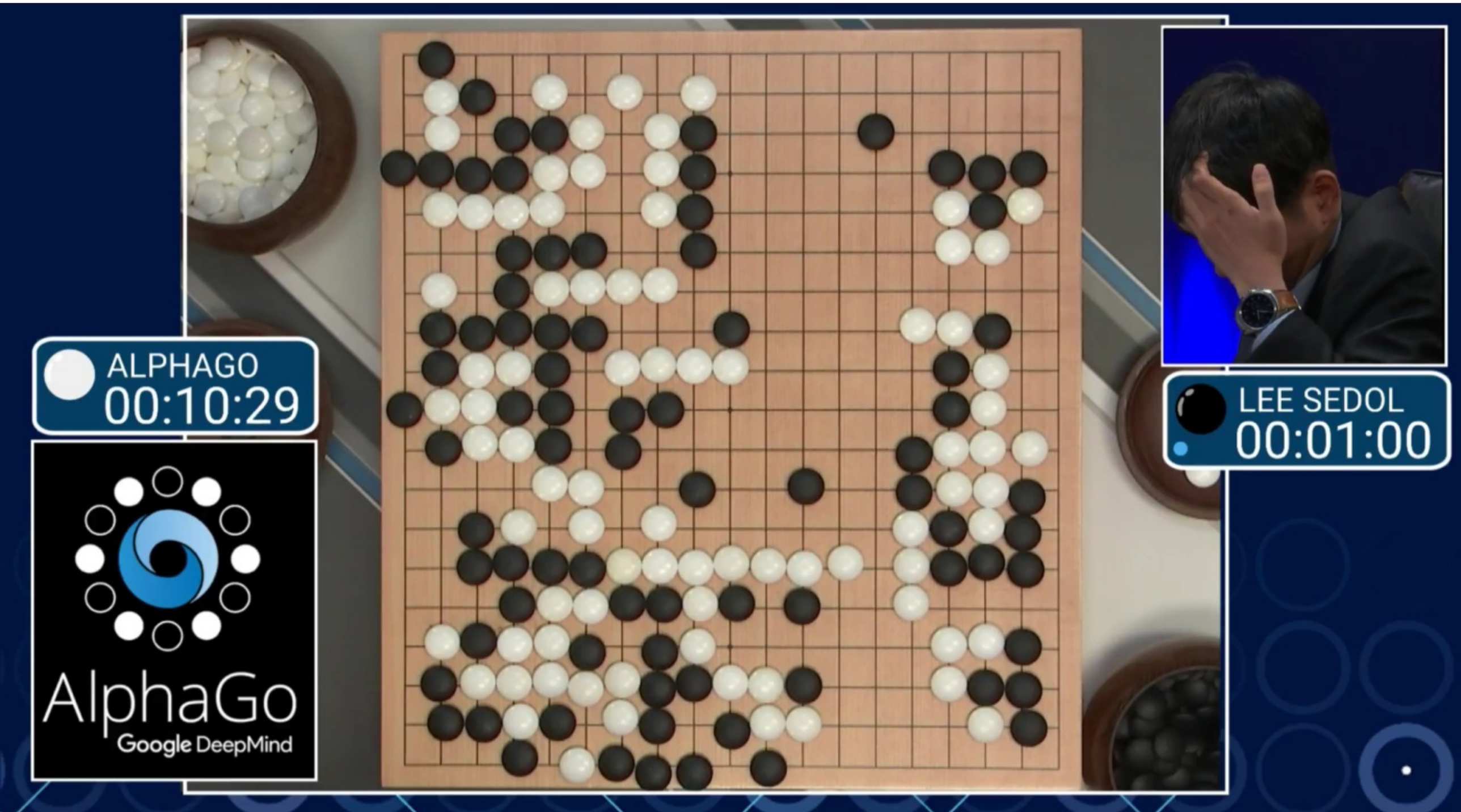
Input
Chest X-Ray Image

CheXNet
121-layer CNN

Output
Pneumonia Positive (85%)



Machine Learning Applications



Machine Learning Gone Wrong

STORYSTREAM TRANSPORTATION UBER RIDE-SHARING

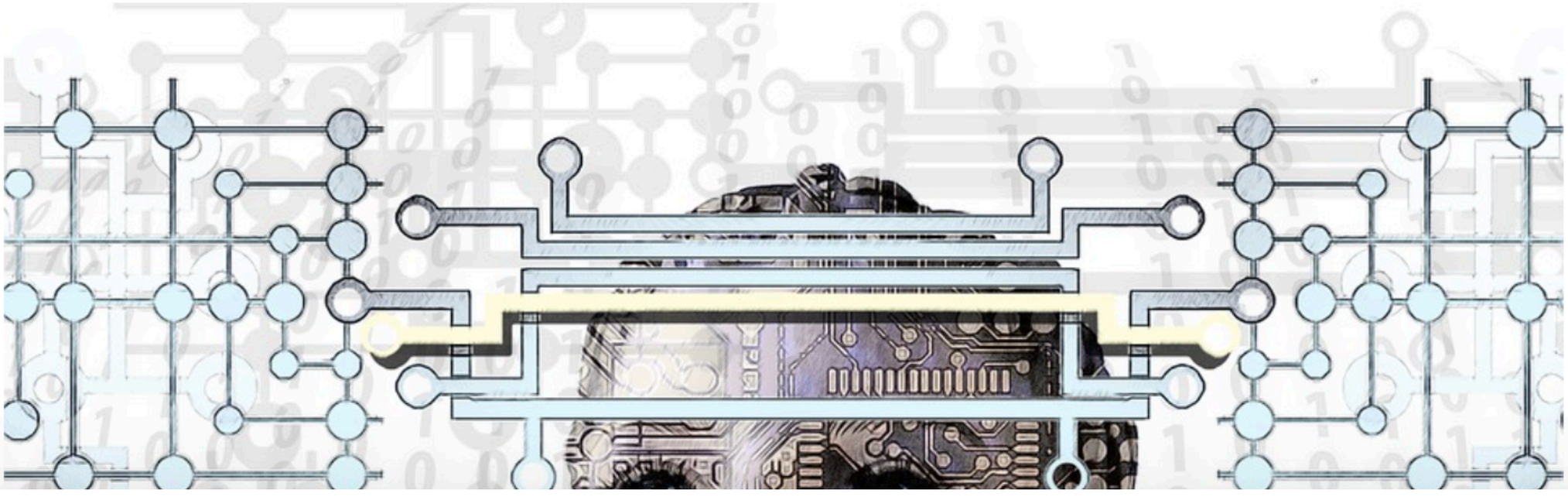
Uber’s fatal self-driving crash: all the news and updates

Machine Learning Gone Wrong

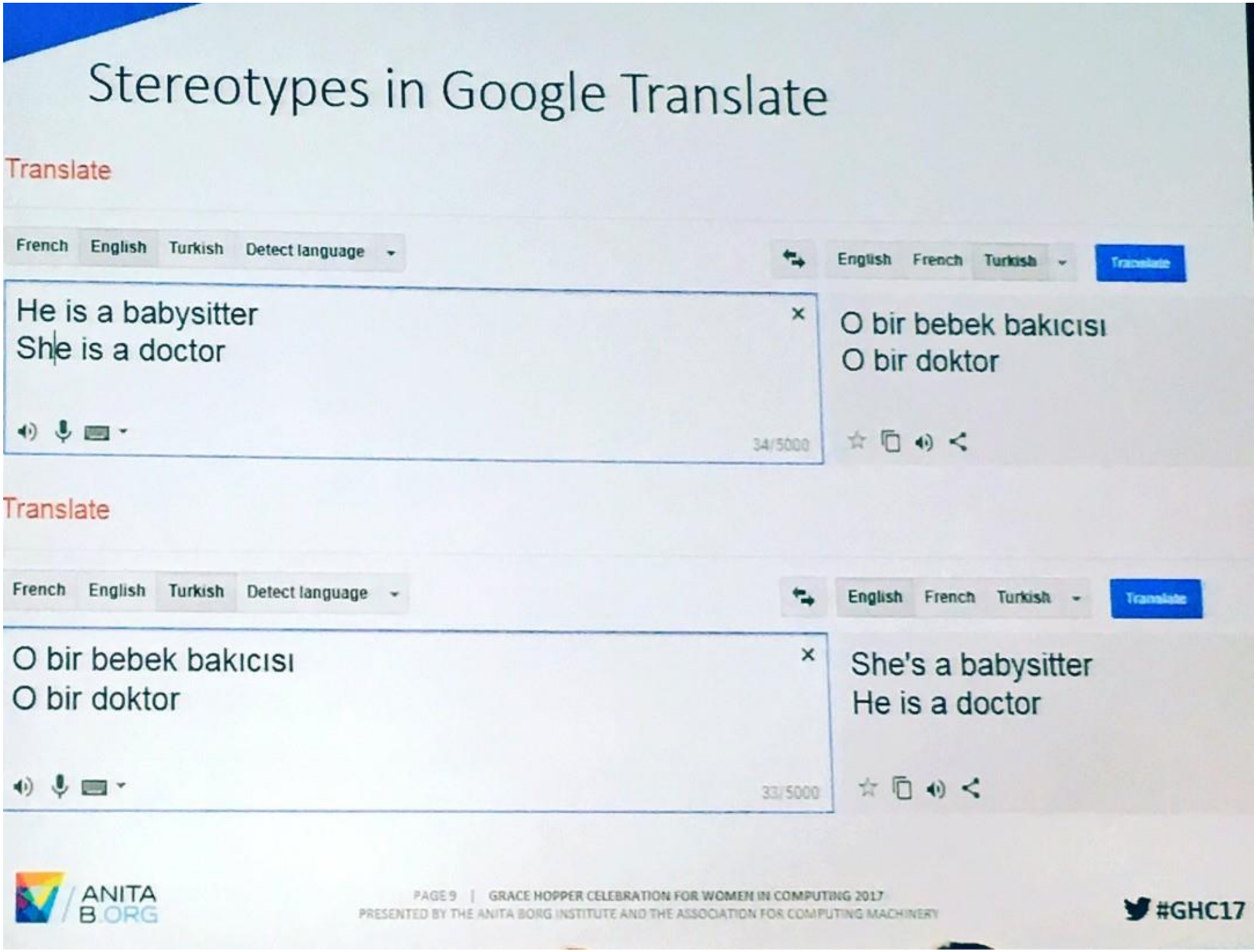
Home › Cool Science › After Uber, Tesla Incidents, Can Artificial Intelligence Be Trusted?

After Uber, Tesla Incidents, Can Artificial Intelligence Be Trusted?

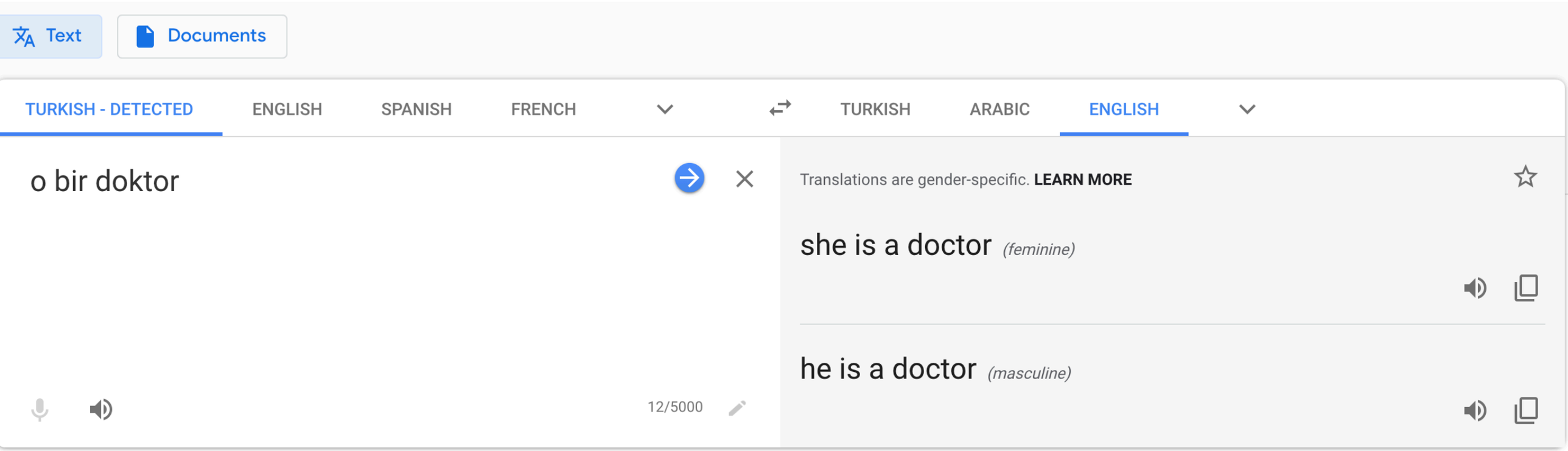
April 13, 2018



“Bias” in Machine Learning

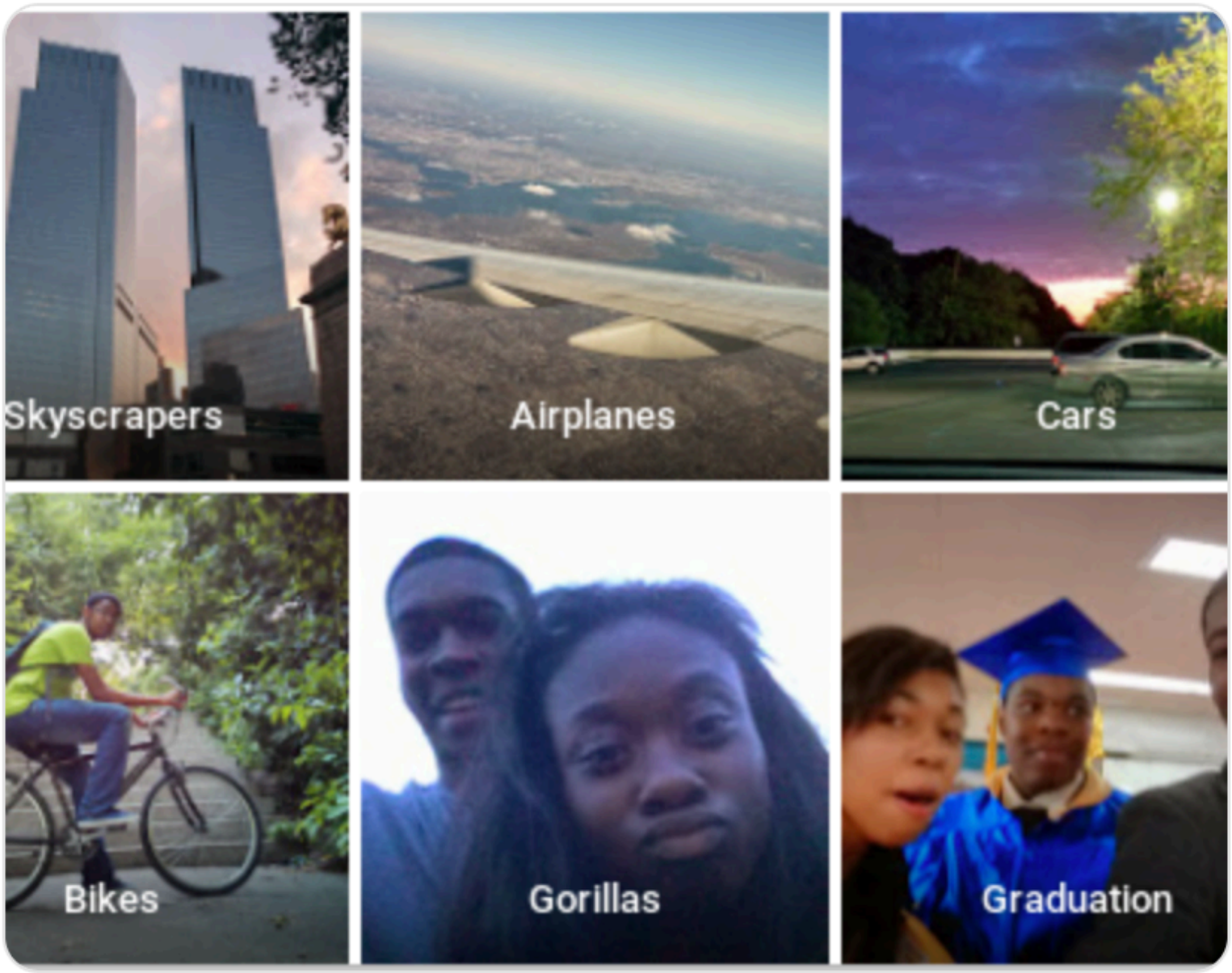


“Bias” addressed



“Racist” Machine Learning?

not a gorilla.



6:22 PM - 28 Jun 2015

Where is the bride?

“Bias” addressed

Machine learning and bias: https://www.youtube.com/watch?v=59bMh59JQDo

“Bias” addressed

Machine learning and bias: https://www.youtube.com/watch?v=59bMh59JQDo

A “reality” check

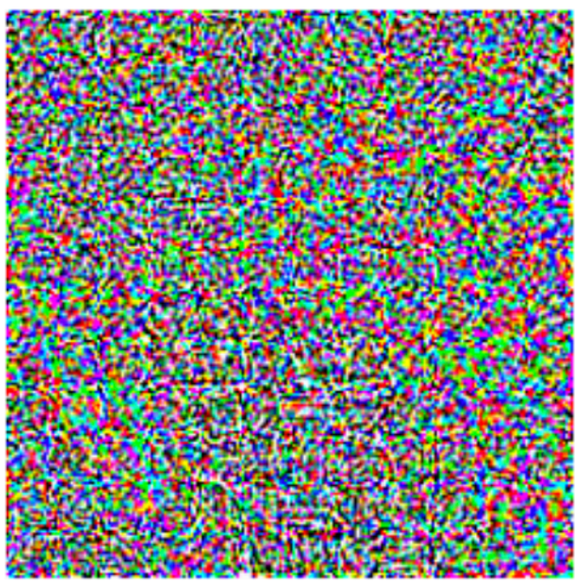
https://www.youtube.com/watch?v=UCwbJxW-ZRg

A “reality” check

https://www.youtube.com/watch?v=UCwbJxW-ZRg

Adversaries!

$+ .007 \times$



$=$

$\boldsymbol{x}$

“panda”
57.7% confidence

$\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$

“nematode”
8.2% confidence

$\boldsymbol{x} +$
$\epsilon \text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
“gibbon”
99.3 % confidence

What is Machine Learning?

What is Machine Learning?

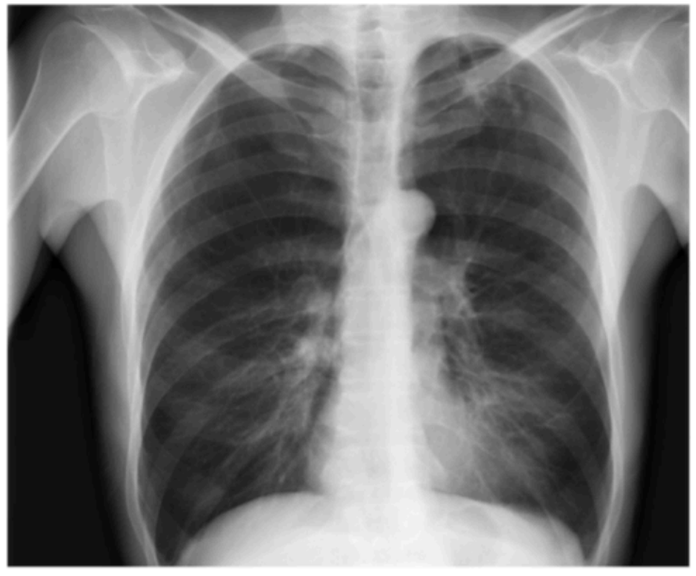
- “Field of study that give computers the ability to learn without being explicitly programmed” - Arthur Samuel [1959]

What is Machine Learning?

What is Machine Learning?

- “A computer program is said to **learn** from **experience E** with respect to some class of **tasks T** and **performance measure P** if its performance at tasks in T, as measured by P, improves with experience E.” - Tom Mitchell

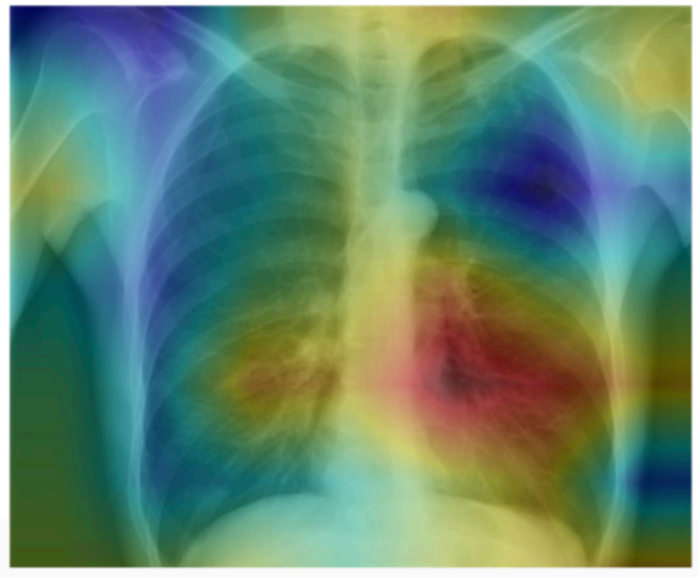
What is Machine Learning?



Input
Chest X-Ray Image

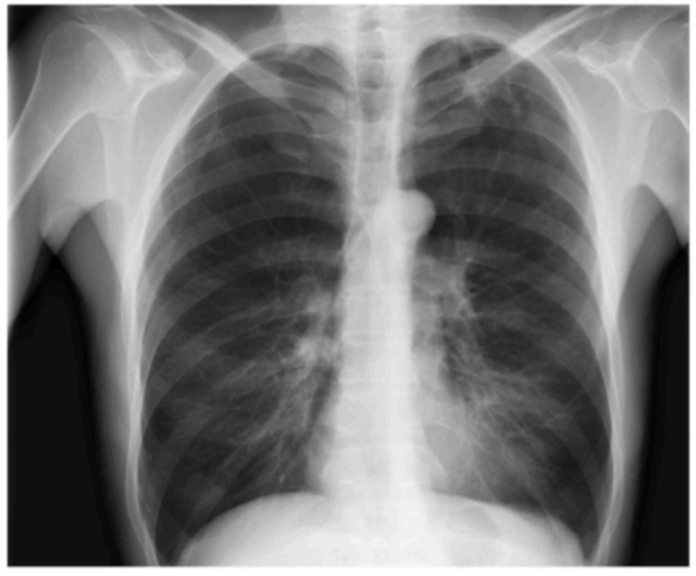
CheXNet
121-layer CNN

Output
Pneumonia Positive (85%)



Q: Identify task, performance measure, and experience

What is Machine Learning?



Input
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Task

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121-layer CNN

Output
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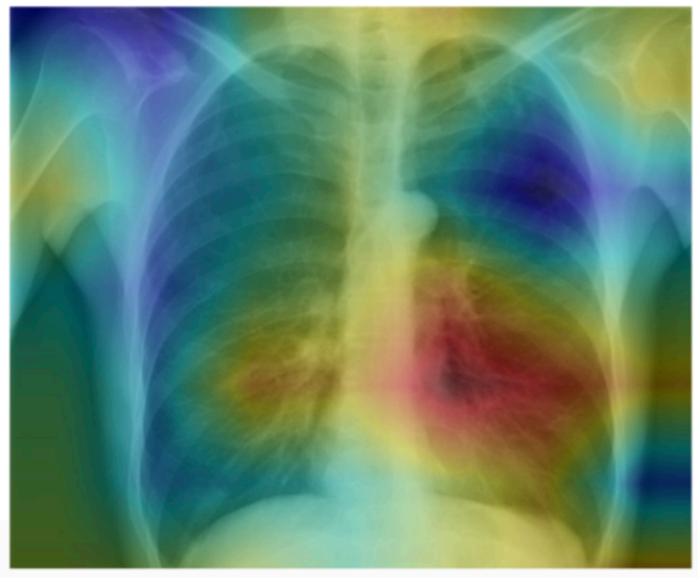

Q: Identify task, performance measure, and experience

What is Machine Learning?

Performance measure

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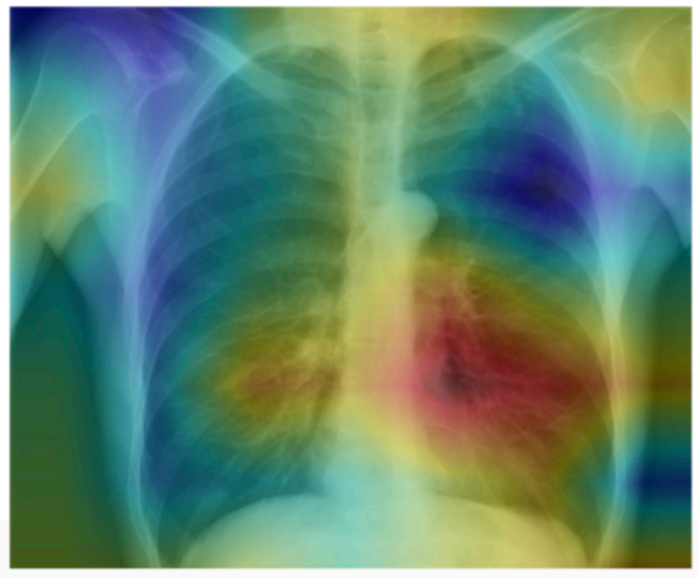

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What is Machine Learning?

Experience
1000s of <image, disease> pairs

Performance measure

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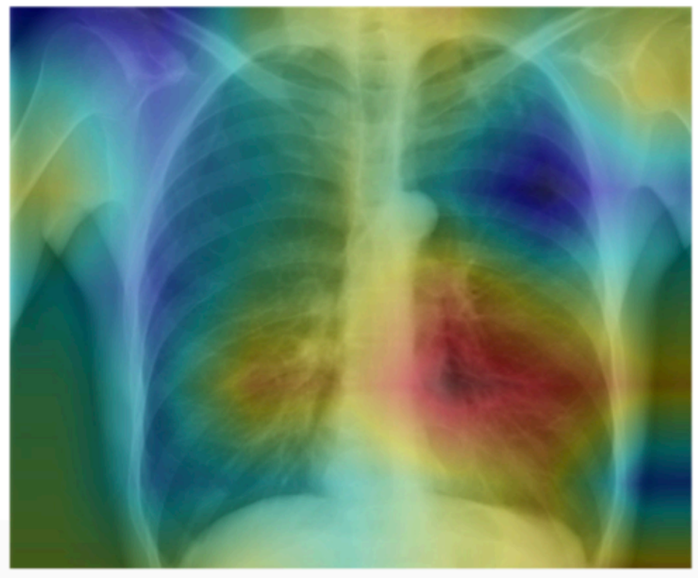

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Performance measure

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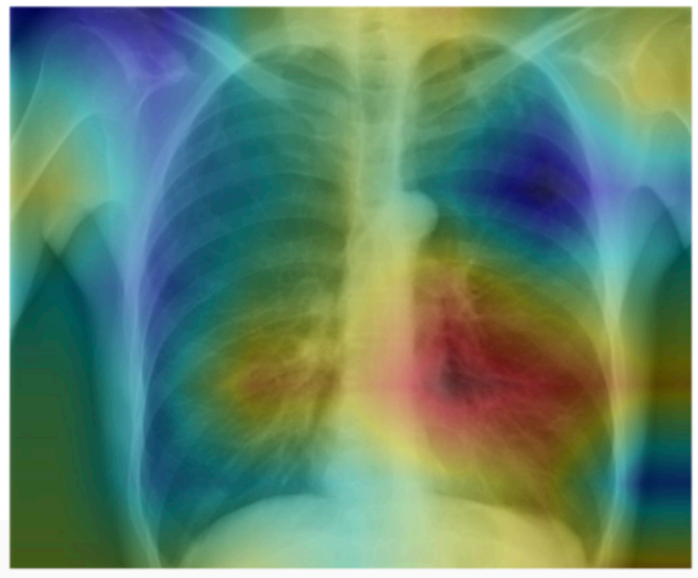

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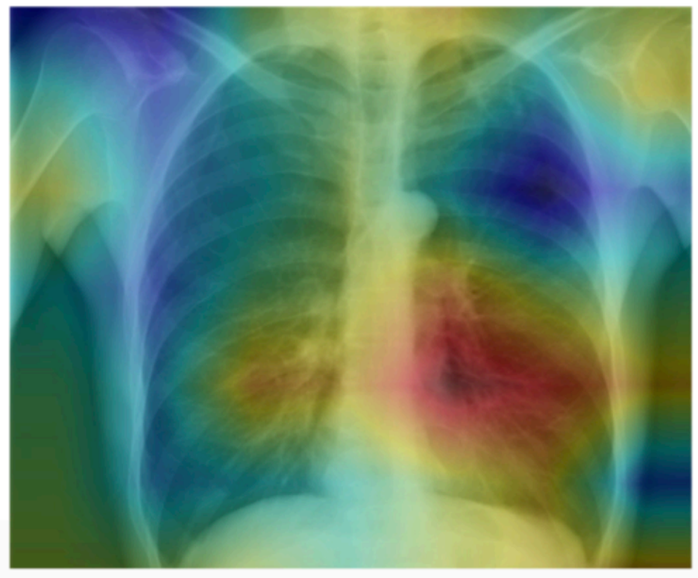

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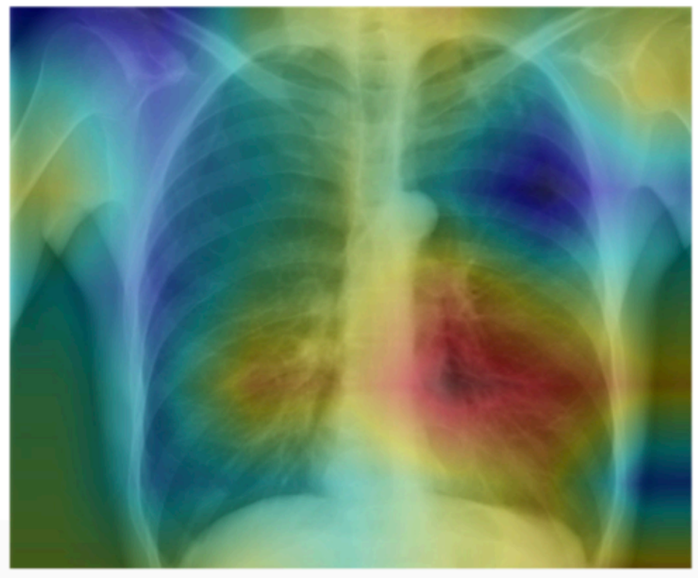

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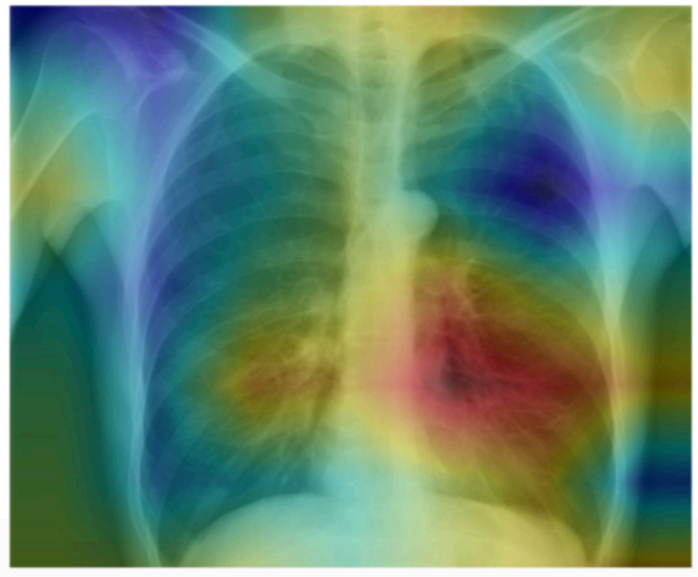

What is Machine Learning?

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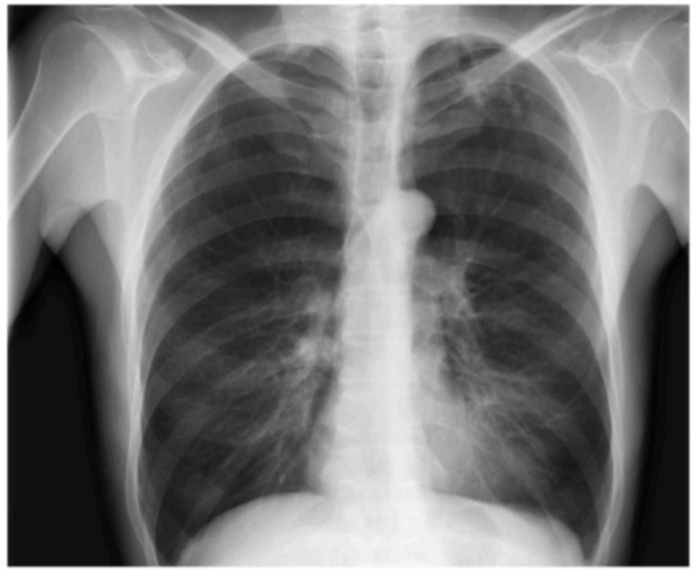
1000s of <image, disease> pairs

Supervised Learning



Input
Chest X-Ray Image

CheXNet
121-layer CNN

Output
Pneumonia Positive (85%)